High-Dimensional Instrumental Variable Quantile Regression under Weak Instruments: A Monte Carlo Study

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Abstract

This thesis studies the finite-sample consequences of weakening instrument relevance in high-dimensional instrumental variable quantile regression (IVQR). Building on a preserved DML-IVQR Monte Carlo design, instrument strength is systematically varied through $\kappa \in \{1, 0.50, 0.25, 0.10\}$ while holding the marginal treatment distribution, structural outcome equation, and control dimension fixed. Three estimators—Oracle-GMM (an infeasible benchmark with known relevant controls), Full-GMM (unregularized estimation with all controls), and DML-IVQR (regularized nuisance estimation with pivotal penalty calibration)—are compared across five quantiles, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and sample sizes $n \in \{500, 1000\}$.

The results show that as κ falls, point-estimation errors (MAE and RMSE) rise sharply, accepted sets expand, and power against false treatment-effect values declines for all estimators, including the oracle benchmark. Coverage is also not always well calibrated in some finite-sample cells, even under strong instrument relevance. DML often improves on Full-GMM but does not uniformly match Oracle-GMM across all design cells. The main conclusion is that Neyman orthogonalization and regularized nuisance estimation can reduce high-dimensional estimation costs, but they cannot replace identifying information lost when instruments become weakly relevant.

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1 Introduction

Economic effects are rarely well summarized by a single average. The impact of a treatment may differ substantially across the outcome distribution, while the treatment itself may be endogenous and the empirical specification may require a large set of controls. Instrumental variable quantile regression (IVQR) provides a framework for studying such heterogeneous structural effects under endogeneity, and double/debiased machine learning (DML) extends this framework to settings with high-dimensional nuisance parameters. However, even with high-dimensional controls and orthogonalization, weak instruments can still remain a serious problem. When identification deteriorates, point estimates can become unstable and confidence regions can lose informativeness even when inference remains formally valid. In this thesis we study that problem directly by examining how progressively weaker instruments affect the finite-sample performance and informativeness of weak-identification-robust DML-IVQR inference, and whether these effects differ across the outcome distribution.

Quantile treatment effects describe how a structural effect varies across quantiles of the potential-outcome distribution. When treatment is endogenous, conventional quantile regression does not generally recover these effects because treatment choices may be related to unobserved determinants of the outcome. IVQR solves this problem by combining excluded instruments with restrictions on the structural ranks of potential outcomes (Chernozhukov and Hansen 2005).

Two additional complications arise in modern applications. First, empirical specifications may contain many controls, transformations, and interactions. Their coefficients are nuisance parameters rather than the primary object of interest, but estimating a large nuisance model without structure can be unstable and may contaminate inference on the treatment effect. Regularization provides one way to estimate such high-dimensional components, while Neyman-orthogonal scores reduce the first-order sensitivity of the target moment to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). Chen et al. (2021) combine these ideas with IVQR and develop a DML-IVQR procedure that handles high-dimensional controls while conducting candidate-specific inference on the structural quantile effect.

Second, regularization does not solve weak identification. Even a valid excluded instrument may contain little information about the endogenous treatment. When instrument-induced treatment variation is small, different candidate treatment-effect values can generate similar population moments, making them difficult to distinguish statistically. This problem is particularly important in IVQR because the relevant identifying restrictions are quantile specific and nonlinear. Standard first-stage measures indicate how strongly the instrument is

related to treatment, but they do not fully capture how much information the instrument provides for identifying a structural quantile effect. The broader weak-instrument literature shows why conventional estimate-centered approximations can become unreliable when identifying information is weak, and why candidate-specific procedures can remain useful even when point identification is imprecise (Staiger and Stock 1997; Stock and Wright 2000; Andrews et al. 2019). In IVQR, this distinction is reflected in weak-identification-robust test inversion: a procedure may remain asymptotically valid while producing broad sets of accepted candidate values and little power against economically relevant false alternatives.

High-dimensional nuisance estimation and weak instrument relevance are conceptually different problems. Neyman orthogonality reduces sensitivity to nuisance-estimation errors; it does not create identifying variation. Weak-identification-robust inference avoids relying on strong local identification, but cannot guarantee precise or informative confidence regions. Performance may therefore break down as instruments weaken even when nuisance estimation is handled well.

This distinction motivates the empirical question studied in this thesis. Existing Monte Carlo evidence for DML-IVQR evaluates regularized nuisance estimation in a high-dimensional design with a fixed level of instrument relevance. It therefore does not systematically show how the same estimators and inference procedures behave when excluded-instrument relevance is progressively weakened while the rest of the structural environment is held fixed. The central research question is: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles?

The thesis answers this question through a controlled extension of the Monte Carlo design in Chen et al. (2021). A single parameter, $\kappa \in \{1.00, 0.50, 0.25, 0.10\}$, controls the loading of the excluded-instrument components in the latent treatment equation. Smaller values reduce excluded-instrument relevance. An independent component is added to the treatment index to preserve its marginal variance as κ varies. The marginal treatment distribution, endogeneity, structural outcome equation, true quantile treatment effects, control dimension, and observed instrument construction are also preserved. This keeps the four relevance settings directly comparable while changing only how strongly the excluded instruments contribute to treatment. The parameter κ is a controlled relevance parameter for this Monte Carlo design, not a universal scalar measure of IVQR identification strength.

Three estimators are compared on identical simulated samples. Oracle-GMM serves as an infeasible benchmark because it knows the true set of relevant controls. Full-GMM uses the full high-dimensional control vector without regularization. DML-IVQR uses regular-

ized nuisance estimation together with the preserved Belloni–Chernozhukov penalty implementation. The comparison separates the common effect of declining instrument information from estimator-specific consequences of nuisance estimation, although differences between Full-GMM and DML-IVQR cannot be attributed to regularization alone because their residualization implementations also differ. Performance is evaluated across five quantiles, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and two sample sizes, $n \in \{500, 1000\}$.

The evaluation separates point-estimation accuracy, coverage, power, and accepted-set informativeness. Bias, mean absolute error (MAE), and root mean squared error (RMSE) summarize accuracy. Coverage measures retention of the true structural quantile effect by the candidate-specific test, while power measures rejection of false treatment-effect values. Inferential informativeness is summarized by the grid-based accepted-set measure within the prespecified primary computational domain $A_0 = [-1, 3]$; the choice of this domain and the corresponding domain-expansion diagnostic are explained in Appendix A. These distinctions matter: high coverage may accompany acceptance of many false candidates, while narrow accepted sets may reject the true value too frequently.

The Monte Carlo evidence shows a break down in informativeness as instrument relevance declines. Point-estimation error rises gradually for Oracle-GMM, Full-GMM, and DML-IVQR; accepted sets expand; and power against false candidate values falls. The deterioration of the oracle benchmark is particularly informative because it shows that the loss is not generated only by high-dimensional nuisance estimation. At the same time, the magnitude of the deterioration varies across quantiles and estimators. Coverage is also not uniformly close to its nominal level in finite samples, including in some strong-relevance cells. These results describe the implemented tests using a chi-square reference cutoff; the experiment does not establish the applicability of the asymptotic regularity conditions or identify the cause of undercoverage. DML-IVQR often performs better than the unregularized high-dimensional benchmark, but it does not dominate in every design cell and cannot compensate for the loss of identifying information when instruments become only weakly relevant. The main contribution of the thesis is therefore to document, within a controlled high-dimensional IVQR experiment, how nuisance robustness, instrument relevance, inferential informativeness, and finite-sample calibration interact as the information supplied by the excluded instruments is systematically reduced.

2 Literature Review

Quantile regression, introduced by Koenker and Bassett (1978), describes how covariates are related to different parts of a conditional outcome distribution rather than only to its mean. This is useful when treatment effects differ across the outcome distribution. Standard quantile regression, however, does not generally recover structural treatment effects when treatment is endogenous, because treatment may be related to unobserved factors that also affect the outcome. Chernozhukov and Hansen (2005) address this problem through instrumental variable quantile regression (IVQR). Their framework combines excluded instruments with restrictions on structural ranks to identify quantile-specific treatment effects in the presence of endogeneity.

A central feature of IVQR is the distinction between instrument validity and identification. Under the structural assumptions of Chernozhukov and Hansen (2005), instrument exogeneity and rank restrictions imply conditional moment conditions that can be used for estimation. Rank invariance assumes that an individual's structural rank remains the same across treatment states, while the weaker rank-similarity condition allows the rank to change as long as its conditional distribution is preserved. These assumptions connect observed outcomes to structural quantiles. They do not guarantee that the structural effect is precisely identified. The instrument must also provide enough treatment variation to distinguish the true structural effect from competing candidate values. Because IVQR moment conditions are quantile specific and nonlinear, conventional first-stage measures provide useful information about instrument relevance but do not fully describe identification at a particular quantile (Chernozhukov and Hansen 2005; Chernozhukov et al. 2017).

Chernozhukov and Hansen (2006) develop practical estimation and inference methods for the IVQR model under regular identification. Their inverse quantile regression approach considers candidate values of the endogenous-treatment coefficient and uses conventional quantile regression to estimate the remaining parameters. This reduces the difficult optimization problem to the lower-dimensional treatment parameter and provides a practical route for IVQR estimation. Chernozhukov and Hansen (2008) extend this approach to inference that is robust to weak identification. Instead of relying on a normal approximation centered on a precisely estimated treatment effect, their procedure tests each candidate treatment-effect value directly and constructs a confidence region by retaining the values that are not rejected. The resulting region can be wide, asymmetric, or disconnected when the instruments contain little identifying information. This is an important distinction for our thesis: weak-identification-robust inference may remain valid while providing little information about the value of the structural

effect. Chernozhukov et al. (2009) develop a different finite-sample conditional inference procedure for quantile regression models.

The IVQR weak-identification problem is related to the broader literature on weak instrumental variables. Anderson and Rubin (1949) provide an early approach based on testing structural parameter values without requiring a precisely estimated first stage. Staiger and Stock (1997) show why conventional instrumental-variable approximations can perform poorly when instruments are weak, while Stock and Wright (2000) develop weak-identification-robust methods for generalized method-of-moments (GMM) models. Kleibergen (2002) and Moreira (2003) provide further robust testing procedures, and Andrews et al. (2019) review the modern weak-instrument literature and its implications for empirical research. These contributions establish the importance of inference procedures that remain reliable when identification is weak. However, the usual linear-IV diagnostics do not provide a direct measure of quantile-specific IVQR identification. For this reason, we report the first-stage statistics to describe the ordering of instrument relevance across the Monte Carlo designs rather than formal IVQR weak-instrument thresholds.

A separate literature addresses inference with many controls or other high-dimensional nuisance parameters. Belloni and Chernozhukov (2011) study ℓ_1-penalized quantile regression under approximate sparsity and develop a pivotal method for choosing the penalty level. This type of regularization makes estimation feasible when the number of possible controls is large, although regularization can itself introduce estimation error. Belloni et al. (2019) show how orthogonal score functions can support valid inference on a parameter of interest in high-dimensional approximately sparse quantile regression models. More generally, Chernozhukov et al. (2018) develop the double/debiased machine learning (DML) framework. Its central idea is to construct Neyman-orthogonal scores whose population moments are less sensitive, to first order, to small errors in estimated nuisance parameters. Additionally, Cross-fitting can reduce overfitting dependence by estimating nuisance functions and evaluating the score on different parts of the sample. Orthogonality and cross-fitting should therefore be distinguished: orthogonality is a property of the score, while cross-fitting is a way of organizing estimation across observations.

These high-dimensional methods address nuisance estimation rather than weak identification. In particular, regularization and Neyman orthogonality cannot make an instrument more relevant or create information that is absent from the data. Chen et al. (2021) combine IVQR with high-dimensional nuisance estimation by developing a DML-IVQR procedure based on a Neyman-orthogonal score. For each candidate treatment-effect value, nuisance parameters are estimated with regularization and the resulting score is used for GMM estima-

tion and candidate-specific inference. Their high-dimensional Monte Carlo design compares Oracle-GMM, which knows the relevant controls, Full-GMM, which uses all controls without regularization, and DML-IVQR, which uses regularized nuisance estimation. They also construct weak-identification-robust confidence regions for these estimators. Their results show that DML-IVQR can substantially improve on an unregularized high-dimensional specification and can approach the oracle benchmark in their baseline designs.

The implementation of Chen et al. (2021) is especially important for our thesis because it provides the baseline Monte Carlo environment that is preserved and extended. Their main high-dimensional simulation considers a fixed level of instrument relevance. It therefore does not systematically examine how the same estimators and candidate-specific inference procedures behave as instrument relevance becomes progressively weaker. This distinction matters because successful high-dimensional nuisance estimation does not imply that the structural parameter remains well identified. A DML procedure may estimate nuisance components accurately while the instruments still provide too little information to distinguish competing structural effects.

Other work illustrates continuing developments in IVQR estimation and high-dimensional quantile methods. Wüthrich (2019) develops an approach for estimating quantile treatment effects with endogenous binary treatment and instruments, providing an alternative to directly optimizing the standard IVQR objective. More recently, Jin and Sun (2025) develop a Neyman-orthogonal moment for a nonparametric IVQR model with high-dimensional covariates and focus on unconditional quantile treatment effects.

The literature therefore leaves a specific question that is central to our thesis. Existing works provide structural IVQR model, weak-identification-robust inference, high-dimensional regularization, and orthogonal-score methods, but there is a limited evidence on how weak instrument relevance affects these components jointly in the preserved high-dimensional DML-IVQR design of Chen et al. (2021). In our thesis we addresse that question by systematically reducing instrument relevance while keeping the remaining features of the Monte Carlo environment as stable as possible. It evaluates how point-estimation accuracy, coverage, power, and accepted-set informativeness change across instrument-relevance settings, quantiles, and estimators. The contribution is therefore a controlled finite-sample extension of the existing DML-IVQR design of Chen et al. (2021).

3 Econometric Framework

3.1 Structural IVQR Model and Identification

Following the IVQR framework of Chernozhukov and Hansen (2005), let Y_i denote the outcome, D_i the endogenous treatment, X_i a vector of observed controls, and Z_i a vector of instruments. The structural quantile function is denoted by $q(u, d, x)$, where $u \in (0, 1)$ represents the structural rank. For a fixed treatment state d and covariates x, $q(\tau, d, x)$ gives the structural τ-quantile of the potential outcome under treatment state d.

Under the IVQR assumptions on instrument exogeneity, treatment selection, and rank similarity, the observed data can be represented as

$$Y_i = q(U_i, D_i, X_i), \qquad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1), \tag{1}$$

where $U_i := U_{D_i i}$ is the realized structural rank. Rank similarity allows treatment-state-specific ranks to differ while requiring their conditional distributions to be the same across treatment states. Rank invariance is the stronger special case in which an individual's rank is unchanged across treatment states. Strict monotonicity of $q(u, d, x)$ in u links the structural rank to the corresponding structural quantile.

Treatment endogeneity arises because the realized treatment D_i may depend on unobserved factors that are related to the structural rank. Consequently, the observed conditional quantile of Y_i given D_i and X_i does not generally equal the structural quantile obtained under a fixed treatment state. IVQR uses instruments to recover the latter object. For two treatment states d_1 and d_0, the structural quantile treatment effect at quantile τ and covariates x is

$$q(\tau, d_1, x) - q(\tau, d_0, x).$$

For a fixed quantile index $\tau \in (0, 1)$, the DML-IVQR framework considered in Chen et al. (2021) uses the linear-in-parameters structural quantile specification

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i' \alpha_0(\tau) + X_i' \beta_0(\tau). \tag{2}$$

The parameter of interest is $\alpha_0(\tau)$, the structural treatment coefficient at quantile τ. The constant term $c_0(\tau)$ and control coefficients $\beta_0(\tau)$ are nuisance parameters. Their specific values in the Monte Carlo data-generating process are introduced later when the simulation design is defined.

The key IVQR restriction follows from the structural-rank representation. Because $q(u, d, x)$ is strictly increasing in u,

$$Y_i \le q(\tau, D_i, X_i) \quad \Longleftrightarrow \quad U_i \le \tau.$$

Since the realized structural rank is conditionally uniform given X_i and Z_i,

$$\Pr(U_i \le \tau \mid X_i, Z_i) = \tau.$$

Substituting the linear structural quantile specification gives

$$\Pr\big[Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau) \mid X_i, Z_i\big] = \tau. \tag{3}$$

Let $\mathbf{1}\{A\}$ denote the indicator function, equal to one when statement A is true and zero otherwise. The same restriction can then be written as the conditional moment condition

$$\mathrm{E}\big[\tau - \mathbf{1}\big\{Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau)\big\} \;\big|\; X_i, Z_i\big] = 0. \tag{4}$$

Equation (4) is the central population restriction used by IVQR. Instrument exogeneity justifies the conditional moment restriction, while identification requires the instruments to provide enough variation to distinguish the true structural coefficient from competing candidate values. These are separate requirements: an instrument may satisfy the validity conditions but still provide too little information for precise identification of $\alpha_0(\tau)$.

3.2 Orthogonal Score and Nuisance Estimation

Fix a quantile τ and consider a candidate value a for the structural treatment coefficient $\alpha_0(\tau)$. IVQR evaluates such candidate values one at a time. For a given a, subtracting $D_i'a$ from the outcome leaves an adjusted outcome whose conditional τ-quantile must be explained by a constant term and the controls. The first nuisance-estimation problem therefore profiles these quantities conditional on the candidate a.

Define the control vector including a constant term as

$$\widetilde{X}_i = (1, X_i')',$$

where the first component is the constant and the remaining components are the p nonconstant controls. Quantile regression measures fit with the check-loss function

$$\rho_\tau(u) = u\big[\tau - \mathbf{1}\{u < 0\}\big]. \tag{5}$$

Here u is a generic scalar residual argument of the check-loss function. The check loss is asymmetric: observations above and below the candidate quantile receive weights determined by τ. Minimizing this loss therefore fits the conditional τ-quantile rather than the conditional mean.

Following Chen et al. (2021), for each candidate a, let $c(a)$ denote the profiled intercept and let $\beta(a)$ denote the profiled vector of control slopes. They are defined jointly by

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg\min_{c,b} \mathrm{E}\big[\rho_\tau\big(Y_i - D_i'a - c - X_i'b\big)\big]. \tag{6}$$

Here $c \in \mathbb{R}$ is the optimization variable for the constant term and $b \in \mathbb{R}^p$ is the optimization vector for the control slopes. Their minimizers are $c(a)$ and $\beta(a)$. These are candidate-specific nuisance parameters: changing a changes the adjusted outcome $Y_i - D_i'a$, and therefore generally changes the fitted constant term and control coefficients. At the true candidate $a = \alpha_0(\tau)$, maintained uniqueness gives $c(a) = c_0(\tau)$ and $\beta(a) = \beta_0(\tau)$.

$$\mathrm{E}\big[\{\tau - \mathbf{1}\big[Y_i \le D_i'a + c(a) + X_i'\beta(a)\big]\}\,\widetilde{X}_i\big] = 0. \tag{7}$$

The moment above is the first-order quantile-regression condition for the profiled constant term and controls. It says that, at the fitted candidate-specific quantile, the quantile residual is orthogonal to the controls together with the constant term.

With the candidate treatment effect and profiled nuisance coefficients fixed, define the candidate residual

$$\epsilon_i(a) = Y_i - D_i'a - c(a) - X_i'\beta(a). \tag{8}$$

The residual equals zero at the candidate fitted quantile. The local behavior of a quantile moment depends on how much probability mass lies near this zero boundary. This is why the residual density at zero enters the orthogonalization. Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ be its conditional density at zero.

Suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\widetilde{X}_i \in \mathbb{R}^{p+1}$. The next step removes from the instruments the component that is locally associated with the controls. Because quantile moments change according to the residual density at zero, this residualization is density weighted rather than an ordinary unweighted linear projection. Define

$$\begin{aligned} M(a) &= \mathrm{E}\big[Z_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\big], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathrm{E}\big[\widetilde{X}_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\big], \quad J(a) \in \mathbb{R}^{(p+1)\times(p+1)}. \end{aligned} \tag{9}$$

Assuming $J(a)$ is nonsingular, set

$$\delta(a) = M(a)J(a)^{-1}, \qquad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \tag{10}$$

Thus $J(a)$ describes density-weighted variation in the controls together with the constant term, while $M(a)$ describes their density-weighted relationship with the instruments. The matrix $\delta(a) = M(a)J(a)^{-1}$ contains the coefficients of the corresponding population residualization. Partition this matrix as $\delta(a) = [d_0(a), \Gamma(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization constant terms and $\Gamma(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\Psi_i(a) = Z_i - \delta(a)\widetilde{X}_i = Z_i - d_0(a) - \Gamma(a)X_i. \tag{11}$$

The vector $\Psi_i(a)$ is therefore the instrument after removing the density-weighted component associated with the constant term and controls. It is this residualized instrument, rather

than the raw Z_i, that enters the theoretical orthogonal score. By construction, $\mathrm{E}[(Z_i - \delta(a)\widetilde{X}_i)\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)] = 0$; this is not an ordinary unweighted residual or a conventional first-stage coefficient.

For the simulation DGP this idealized population construction requires a qualification. At the true structural parameter, write $e_\tau = D(u - q_\tau)$, where $q_\tau = \Phi^{-1}(\tau)$ and $D = \Phi(T)$ with $T = d^*(\kappa)$. Conditional on $u = q_\tau$, X, and the observed instruments Z, T is normal with variance $2.91 - \kappa^2 > 1$ for every investigated κ. The population residual density at zero contains a term proportional to $\mathrm{E}[1/\Phi(T) \mid u = q_\tau, X, Z]$, which is not finite under these conditional normal variances.

The candidate-specific nuisance vector collects the quantities required to construct the score.

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \mathrm{vec}\{\delta(a)\} \end{pmatrix}, \qquad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \mathrm{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \tag{12}$$

Vectorization only stacks the residualization matrix. The target remains $\alpha_0(\tau)$; $c(a)$ and $\beta(a)$ profile the outcome nuisance, while $\delta(a)$ residualizes the instruments.

Following Chen et al. (2021), the orthogonal score combines two objects: the candidate quantile residual and the density-weighted residualized instrument.

$$\begin{aligned} g_\tau(W_i; a, \eta) &= \left[\tau - \mathbf{1}\{Y_i \le D_i' a + c + X_i'\beta\}\right]\left(Z_i - \delta\widetilde{X}_i\right), \\ g_\tau(W_i; a, \eta(a)) &= \left[\tau - \mathbf{1}\{Y_i \le D_i' a + c(a) + X_i'\beta(a)\}\right]\Psi_i(a), \end{aligned} \tag{13}$$

where $W_i = (Y_i, D_i, X_i, Z_i)$. The first factor asks whether the observed outcome lies below the candidate structural quantile. The second factor supplies instrument variation after removing the control-related component. At the true treatment coefficient and true nuisance values, their population product has expectation zero.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$\mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \tag{14}$$

Ordinary estimation of high-dimensional nuisances can affect a target moment. DML therefore uses a score with the stronger local property that its population expectation is insensitive, to first order, to small changes of the nuisance vector at the truth. Formally,

$$\partial_\eta \, \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} = 0. \tag{15}$$

Here ∂_η denotes differentiation with respect to the nuisance vector. Orthogonality does not mean that nuisance estimation is error free. It means that sufficiently small nuisance-estimation errors do not enter the target moment at first order.

The cancellation can be seen directly. First consider changes of the profiled constant term and control slopes. To show the cancellation for the constant term and slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$ and define

$$\begin{aligned} M_0 &= \mathrm{E}\big[Z_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ J_0 &= \mathrm{E}\big[\widetilde{X}_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \tag{16}$$

At the truth, differentiation with respect to the fitted constant term and control slopes then gives

$$\begin{aligned} \partial_{(c,\beta')} \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} &= -\mathrm{E}\big[(Z_i - \delta_0 \widetilde{X}_i)\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big] \\ &= -\left[M_0 - \delta_0 J_0\right] = 0. \end{aligned} \tag{17}$$

The density appears because observations near the boundary can switch sides. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\widetilde{X}_i$, cancelling first-order changes in the constant term and slopes.

A similar argument applies to the instrument-residualization nuisance. For instrument component $j = 1, \ldots, d_z$, let δ_j denote the corresponding row of δ. Then

$$\partial_{\delta_j} \mathrm{E}\big[g_{\tau,j}(W_i; \alpha_0(\tau), \eta)\big]\big|_{\eta=\eta_0} = -\mathrm{E}\big[\big\{\tau - \mathbf{1}\big[Y_i \le D_i'\alpha_0(\tau) + c_0(\tau) + X_i'\beta_0(\tau)\big]\big\}\widetilde{X}_i'\big] = 0. \tag{18}$$

An error in δ is multiplied by the quantile residual, whose moment with $\widetilde{X}_i$ is zero by (7), so its first-order effect also vanishes. This construction permits sufficiently accurate regularized nuisances to support inference on the low-dimensional target (Chernozhukov et al. 2018; Chen et al. 2021).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a. IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.3 Estimators and Executable Implementation

In this thesis, we use the Belloni–Chernozhukov pivotal penalty to implement the quantile-Lasso step of DML-IVQR. Chen et al. (2021, Table 3) compare the Belloni–Chernozhukov penalty rule with five-fold cross-validation and report similar finite-sample per-

formance in terms of RMSE, MAE, and bias in their simulation design. Motivated by this evidence and the additional computational cost of cross-validation, this thesis adopts the Belloni–Chernozhukov rule for selecting the quantile-regression penalty. Section 3.2 introduced the population nuisance parameters. When the number of controls is large relative to n, DML-IVQR estimates their corresponding sample quantities using ℓ_1 regularization under approximate sparsity. This assumes that only a relatively small subset of controls has important predictive information, while the remaining coefficients are zero or small.

Fix τ and a candidate value a. Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nuisance step is $Y_i - D_i'a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \ldots, b_p)'$ contain only the slopes on X_i. Thus, $(\widehat{b}_0(a), \widehat{\beta}(a))$ estimates $(c(a), \beta(a))$. The implementation-specific notation for the later instrument regression is defined below and is kept distinct from the full population matrix $\delta(a) = [d_0(a), \Gamma(a)]$. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \underset{b_0, b}{\arg\min} \left\{ \sum_{i=1}^{n} \rho_\tau\big(Y_i - D_i'a - b_0 - X_i'b\big) + \frac{1}{2}\left[\lambda_0^{rq}(a)|b_0| + \sum_{k=1}^{p} \lambda_k^{rq}(a)|b_k| \right] \right\}. \quad (19)$$

The first term measures fit at quantile τ. The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum. The factor $1/2$ reflects the augmented-observation implementation of `rq.fit.lasso` in the preserved `quantreg` 5.34 environment. Accordingly, $\lambda^{rq}(a)$ denotes the vector supplied to the routine rather than the coefficient multiplying the absolute-value penalty directly. The implementation uses the pivotal quantile-Lasso calibration developed by Belloni and Chernozhukov (2011), following the executable route preserved by Chen et al. (2021). Let $s_k = (n^{-1}\sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \ldots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The vector supplied to the routine is

$$\begin{aligned} T_r &= \max_{1 \le k \le p} \left| \frac{\sum_{i=1}^{n} X_{ik}\,[\tau - \mathbf{1}\{U_{ir} < \tau\}]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda^{rq}(a) &= 2\,\widehat{Q}_{0.9}\Big(\{T_r\}_{r=1}^{1000}\Big)\sqrt{\tau(1-\tau)}(1, s_1, \ldots, s_p)'. \end{aligned} \quad (20)$$

Here $k = 1, \ldots, p$ indexes the control variables, $r = 1, \ldots, 1000$ indexes the simulated pivotal draws, T_r is the simulated pivotal statistic in draw r, and $\widehat{Q}_{0.9}(\{T_r\}_{r=1}^{1000})$ denotes their empirical 90th percentile. The $p + 1$ entries of $\lambda^{rq}(a)$ correspond respectively to the intercept and the p control slopes. Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2

scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned}\widehat{\epsilon}_i(a) &= Y_i - D_i'a - \widehat{b}_0(a) - X_i'\widehat{\beta}(a),\\ \widehat{\mu}_\epsilon(a) &= \frac{1}{n}\sum_{i=1}^{n}\widehat{\epsilon}_i(a),\\ \widehat{f}_i(a) &= \phi\Big(\widehat{\epsilon}_i(a); \widehat{\mu}_\epsilon(a), \widehat{\sigma}^2_\epsilon(a)\Big),\end{aligned} \tag{21}$$

Here $\widehat{\mu}_\epsilon(a)$ denotes the sample mean of the n estimated candidate residuals, and $\widehat{\sigma}^2_\epsilon(a)$ denotes their sample variance. The function $\phi(e;\mu,\sigma)$ is the normal density with mean μ and standard deviation σ. The executable implementation uses these quantities to form a fitted-normal density proxy at the candidate residuals. This proxy is implementation specific and should not be interpreted as a direct estimate of the theoretical conditional density $f_{\epsilon(a)}(0 \mid X_i, Z_i)$. Further implementation details are documented in Appendix D. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component $j = 1, \ldots, d_z$, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{r}_{j0}(a)$ denote its retained ordinary intercept and let $\widehat{\Gamma}_{\text{imp},j}(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix}\widehat{r}_{j0}(a)\\ \widehat{\Gamma}_{\text{imp},j}(a)\end{pmatrix} = \underset{(r_{j0},\gamma_j)}{\arg\min}\left\{\sum_{i=1}^{n}\left[\sqrt{\widehat{f}_i(a)}Z_{ji} - r_{j0} - \sqrt{\widehat{f}_i(a)}X_i'\gamma_j\right]^2 + \mathcal{P}_{j,a}(\gamma_j)\right\}. \tag{22}$$

Here $\mathcal{P}_{j,a}(\gamma_j) = \sum_{k=1}^{p}\lambda^{\Gamma}_{jk}(a)|\gamma_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. Here $\gamma_j \in \mathbb{R}^p$ is the slope vector for instrument component j, and $k = 1, \ldots, p$ indexes its control coefficients. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the intercept and the slopes. Define $\widehat{r}_0(a) = (\widehat{r}_{10}(a), \ldots, \widehat{r}_{d_z0}(a))'$ and stack the slope rows in $\widehat{\Gamma}_{\text{imp}}(a)$. The residualized instrument used in the score is

$$\begin{aligned}\widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{r}_{j0}(a) - X_i'\widehat{\Gamma}_{\text{imp},j}(a),\\ \widehat{\Psi}_i(a) &= Z_i - \widehat{r}_0(a) - \widehat{\Gamma}_{\text{imp}}(a)X_i.\end{aligned} \tag{23}$$

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained constant term and the fitted contribution of the original controls. It does not divide the retained constant term by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The constant term retained by the transformed regression is specific to the implementation. It enters as an ordinary constant rather than as $\sqrt{\widehat{f}_i(a)}$ multiplied by a constant regressor. Therefore, $\widehat{r}_0(a)$ is not the exact sample version of $d_0(a)$ in the population projection of Section 3.2, which includes both the constant term and the controls. The exact derivative cancellation derived in Section 3.2 applies to the corresponding theoretical score. The implemented score follows the density-weighted residualization of the control slopes but handles the constant term differently. The results therefore describe the preserved implementation rather than an exact sample implementation of every part of the theoretical score.

The penalties can shrink weak coefficients to zero and stabilize both high-dimensional steps, but also create shrinkage error. The theoretical score in Section 3.2 limits its first-order effect (Chernozhukov et al. 2018). Nuisances and scores use the same sample; this preserved implementation has no sample splitting or cross-fitting.

Oracle-GMM, Full-GMM, and DML-IVQR target the same $\alpha_0(\tau)$ using the same outcome, treatment, instruments, and quantile. Each multiplies a candidate quantile score by residualized instruments, producing a moment with dimension d_z; nuisance estimation differs.

Oracle-GMM is an infeasible benchmark because it knows the exact ten controls used in the simulation DGP. Let $S_0 = \{1, \ldots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\mathrm{orc}}(a) \\ \widehat{\beta}_{\mathrm{orc}}(a) \end{pmatrix} = \underset{b_0,b}{\arg\min} \sum_{i=1}^{n} \rho_\tau\left(Y_i - D_i'a - b_0 - X_{S_0,i}'b\right). \tag{24}$$

Thus, Oracle-GMM first removes $D_i'a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. Oracle serves as a benchmark to measure the effect of knowing the true oracle set.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\mathrm{full}}(a) \\ \widehat{\beta}_{\mathrm{full}}(a) \end{pmatrix} = \underset{b_0,b}{\arg\min} \sum_{i=1}^{n} \rho_\tau\left(Y_i - D_i'a - b_0 - X_i'b\right). \tag{25}$$

Thus all 100 controls enter freely. Full-GMM does not require the unknown oracle set, but estimating many weak or irrelevant coefficients can be noisy.

Oracle-GMM and Full-GMM use the same unregularized instrument residualization with their respective control matrices. For $e \in \{\mathrm{orc}, \mathrm{full}\}$, let $X_{\mathrm{orc},i} = X_{S_0,i}$ and $X_{\mathrm{full},i} = X_i$, and let

$\widehat{f}_{e,i}(a)$ be the fitted-normal proxy defined above. The code computes

$$\begin{aligned} \widehat{M}_e(a) &= \frac{1}{n}\sum_{i=1}^{n} Z_i X'_{e,i}\widehat{f}_{e,i}(a), \qquad \widehat{J}_e(a) = \frac{1}{n}\sum_{i=1}^{n} X_{e,i}X'_{e,i}\widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a)\widehat{J}_e(a)^{-1}, \qquad \widehat{\Psi}_{e,i}(a) = Z_i - \widehat{\delta}_e(a)X_{e,i}. \end{aligned} \tag{26}$$

This direct, unpenalized inversion includes no residualization constant term, although the quantile regression has one, and uses the original Z_i in the final residual. It therefore differs from the population orthogonalization including the constant term and controls in Section 3.2. The implementation directly inverts $\widehat{J}_e(a)$ without additional regularization or a generalized inverse. If the matrix is singular, the computation is recorded as a numerical failure. Each benchmark multiplies $\widehat{\Psi}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR uses all controls and regularizes both nuisance steps. It estimates the control coefficients by quantile Lasso, forms the density proxy, and uses regularized weighted instrument regressions. DML-IVQR applies (20). Oracle and Full also use residualized IVQR moments, but estimate nuisances without ℓ_1 regularization. Unlike their matrix calculation, DML retains a residualization constant term. The comparison covers a known oracle set without regularization, all controls without regularization, and all controls with regularized nuisance estimation. It does not isolate every implementation difference but keeps the structural model, target, and instruments common.

3.4 GMM Criterion and Weak-Identification-Robust Inference

Let $e \in \{\text{orc}, \text{full}, \text{dml}\}$ index the estimator and $\widehat{g}_{e,i}(a) \in \mathbb{R}^{d_z}$ its observation-level score. Its sample average is

$$\widehat{g}_{e,n}(a) = \frac{1}{n}\sum_{i=1}^{n} \widehat{g}_{e,i}(a). \tag{27}$$

This vector measures the sample restriction at a. The code measures its variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{n}\sum_{i=1}^{n} \widehat{g}_{e,i}(a)\widehat{g}_{e,i}(a)'. \tag{28}$$

This matrix standardizes differently scaled and correlated components. It uses denominator n, no finite-sample correction, and no sample-mean centering. At the truth, its probability limit equals the score covariance because the population moment is zero.

The common candidate-specific GMM criterion is

$$W_{e,n}(a) = n\widehat{g}_{e,n}(a)'\widehat{\Sigma}_e(a)^{-1}\widehat{g}_{e,n}(a). \tag{29}$$

This equals the code's calculation from the unnormalized score sum and sum of outer products. Smaller values indicate moments closer to zero after standardization. The implementation directly inverts the matrix without additional regularization or a generalized inverse. If the inversion fails or produces a non-finite result, the computation is recorded as a numerical failure. Suppressing e gives $W_n(a)$. Let $\mathcal{A}$ denote the set of candidate treatment-effect values over which the GMM profile is evaluated; its finite numerical specification in the Monte Carlo experiment is given in Section 4.4. The point estimator is

$$\widehat{\alpha}_e(\tau) = \arg\min_{a \in \mathcal{A}} W_{e,n}(a). \tag{30}$$

The minimum need not be zero. The implementation minimizes over the finite profile in Section 4.4, selects the first exact tie, and gives no point estimate to an incomplete profile. Point estimators are selected on the fixed numerical grid; except at quantiles whose true effect lies exactly on a grid point, reported point-estimation errors therefore also include a small grid-approximation error.

Point estimation retains the minimizer; inference tests each a separately. Under the regularity conditions for the corresponding theoretical candidate-specific IVQR score, the null statistic has the asymptotic reference distribution

$$W_{e,n}(a) \xrightarrow{d} \chi^2_{d_z}, \qquad d_z = \dim(Z_i). \tag{31}$$

The degrees of freedom equal the number of moment components. The preserved Monte Carlo implementation applies this chi-square law as the asymptotic reference distribution for the candidate-specific statistics of all three estimators. For DML-IVQR, however, the implementation-specific residualization constant term described in Section 3.3 is not the exact sample version of the constant term in the theoretical projection including the controls. Following Chernozhukov and Hansen (2008), the test evaluates the hypothesized value directly rather than using a normal approximation for $\widehat{\alpha}_e(\tau)$. At level γ, reject when $W_{e,n}(a) > c_{1-\gamma,d_z}$. Here $\gamma \in (0,1)$ is the test significance level, and $c_{1-\gamma,d_z}$ is the $(1-\gamma)$-quantile of a chi-square distribution with d_z degrees of freedom. This test-level γ is distinct from the instrument-regression slope vector γ_j used in Section 3.3.

Inverting these candidate-specific tests gives

$$C_{e,1-\gamma}(\tau) = \left\{ a \in \mathcal{A} : W_{e,n}(a) \le c_{1-\gamma,d_z} \right\}. \tag{32}$$

Test inversion retains every candidate not rejected by its own moment test. Because the profile can cross the cutoff repeatedly, the region may be connected, disconnected, wide, or empty. A finite numerical domain describes only acceptance within that domain and cannot establish global unboundedness.

The underlying Chernozhukov–Hansen test-inversion approach can retain asymptotic validity under weak, partial, or failed identification, subject to its regularity conditions, but need not be informative. Weak instruments can leave many candidates below the cutoff. Because the implementation uses an asymptotic chi-square critical value, finite-sample calibration remains empirical. This differs from the exact conditional simulation method of Chernozhukov et al. (2009).

4 Monte Carlo Design

4.1 Baseline Data-Generating Process

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. The baseline design combines endogenous treatment, high-dimensional controls, and two instruments. For each observation i, the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix} \right). \tag{33}$$

so that both disturbances have unit variance and $\mathrm{Cov}(u_i, \epsilon_i) = 0.3$.

The remaining primitive variables are mutually independent and are also independent of the disturbance pair in Equation (33):

$$x_{ji} \overset{\text{ind}}{\sim} \mathcal{N}(0,1), \quad j = 1, \ldots, 100, \qquad z_{1i}, z_{2i}, v_{1i}, v_{2i} \overset{\text{ind}}{\sim} \mathcal{N}(0,1). \tag{34}$$

Observations are independent across i. The observed controls are obtained by applying the standard-normal cumulative distribution function to the latent control variables:

$$X_{ji} = \Phi(x_{ji}), \qquad j = 1, \ldots, 100, \tag{35}$$

Thus $X_{ji} \sim \mathcal{U}(0,1)$, and the observed control vector is $X_i = (X_{1i}, \ldots, X_{100i})'$.

The executable design constructs two observed instruments as[1]

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \tag{36}$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \tag{37}$$

1. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j. The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in our thesis follow that executable implementation.

Let $Z_i = (Z_{1i}, Z_{2i})'$, so the number of instrument components is $d_z = 2$. The excluded variables z_{1i} and z_{2i} provide the part of the instruments that enters the treatment index directly. The v components add independent noise, while the included X terms generate dependence between the instruments and observed controls. Because these primitives are independent of the structural disturbances, their inclusion does not create the treatment endogeneity described below.

In the baseline design, treatment is generated by

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \tag{38}$$

The treatment is continuous and lies in $(0, 1)$. The structural outcome equation is

$$Y_i = 1 + D_i + 5 \sum_{j=1}^{7} X_{ji} + D_i u_i. \tag{39}$$

The outcome depends directly on $X_{1i}, \ldots, X_{7i}$. Together with the controls entering the instruments, $X_{1i}, \ldots, X_{10i}$ form the ten-control oracle set used by Oracle-GMM.

The source of treatment endogeneity can now be seen directly from Equations (33), (38), and (39). The disturbance ϵ_i enters the treatment index, while u_i enters the structural outcome through the interaction $D_i u_i$. Because u_i and ϵ_i are positively correlated, realizations of the treatment are statistically related to the unobserved component of the structural treatment effect. Joint normality gives

$$E[u_i \mid \epsilon_i] = 0.3\, \epsilon_i.$$

Since

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i)$$

is increasing in ϵ_i, this implies

$$\mathrm{Cov}(u_i, D_i) = 0.3\, E[\epsilon_i \Phi(z_{1i} + z_{2i} + \epsilon_i)] > 0.$$

Hence $E[u_i \mid D_i]$ is not generally zero: treatment is endogenous because it is related to an unobserved determinant of the structural response to treatment.

For a fixed treatment value $d \in (0, 1)$, Equation (39) implies the potential outcome

$$Y_i(d) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d(1 + u_i). \tag{40}$$

Because u_i is standard normal and independent of X_i, the conditional structural quantile is

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d\left[1 + \Phi^{-1}(\tau)\right]. \tag{41}$$

where Φ^{-1} denotes the standard-normal quantile function. The structural quantile treatment effect is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \tag{42}$$

For the five quantiles considered in the simulations, the corresponding true structural quantile treatment effects are approximately

$$\alpha_0(0.10) = -0.282, \qquad \alpha_0(0.25) = 0.326, \qquad \alpha_0(0.50) = 1.000,$$
$$\alpha_0(0.75) = 1.674, \qquad \alpha_0(0.90) = 2.282.$$

These values serve as the true treatment effects when evaluating point-estimation accuracy, coverage, and power.

4.2 Instrument Relevance and the κ-Extension

The baseline design of Chen et al. (2021) holds instrument relevance fixed. We introduce a single parameter κ that reduces the impact of the excluded variables z_{1i} and z_{2i} to treatment. Simply replacing $z_{1i} + z_{2i}$ with $\kappa(z_{1i} + z_{2i})$ would also reduce the variance of the latent treatment index. Because treatment is obtained by applying Φ to that index, this would change the marginal treatment distribution together with instrument relevance.

To avoid this additional change, the design introduces an independent standard-normal variable w_i that replaces the variance removed from the excluded-instrument component. Let $w_i \sim \mathcal{N}(0, 1)$ be independent of all previously defined variables. For $\kappa \in [0, 1]$, define

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)}\, w_i + \epsilon_i, \tag{43}$$

$$D_i(\kappa) = \Phi\big(d_i^*(\kappa)\big)\,. \tag{44}$$

The coefficient $\sqrt{2(1 - \kappa^2)}$ preserves the variance contributed by the excluded part of the treatment index. Appendix B derives this coefficient and establishes the distributional properties of the construction. Since

$$\mathrm{Var}(z_{1i} + z_{2i}) = 2,$$

the scaled excluded-instrument component contributes $2\kappa^2$, while the independent w_i component contributes $2(1 - \kappa^2)$. Their combined variance therefore remains equal to two.

At $\kappa = 1$, the compensating component disappears and the baseline treatment equation is recovered:

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \tag{45}$$

As κ falls, the direct contribution of z_{1i} and z_{2i} to treatment becomes smaller. At the conceptual limit $\kappa = 0$, these excluded components disappear from the treatment index. The simu-

lations use positive values down to $\kappa = 0.10$, so the weakest design represents very low but nonzero designed relevance rather than a formal case of non-identification.

The variance of the complete latent treatment index remains constant:

$$\mathrm{Var}\left[d_i^*(\kappa)\right] = 2\kappa^2 + 2(1-\kappa^2) + 1 = 3. \tag{46}$$

Hence $d_i^*(\kappa) \sim \mathcal{N}(0,3)$ for every κ. Since $D_i(\kappa) = \Phi(d_i^*(\kappa))$, the marginal treatment distribution is unchanged across the relevance designs.

Instrument relevance, however, changes directly with κ. From the construction of the observed instruments and the latent treatment index,

$$\mathrm{Cov}\left(Z_{1i}, d_i^*(\kappa)\right) = \mathrm{Cov}\left(Z_{2i}, d_i^*(\kappa)\right) = \kappa. \tag{47}$$

so the instrument–index covariance declines one-for-one with κ. Thus κ provides a transparent design parameter for ordering instrument relevance in this Monte Carlo experiment. It should not be interpreted as a universal measure of IVQR identification strength or as a conventional weak-instrument statistic. The first-stage F-statistic and partial R^2 reported in Chapter 5 are therefore used only as descriptive diagnostics of the relevance ordering; no Stock–Yogo or $F < 10$ threshold is imposed on IVQR.

The relationship generating treatment endogeneity is also preserved. Because u_i is independent of z_{1i}, z_{2i}, and w_i, while $\mathrm{Cov}(u_i, \epsilon_i) = 0.3$,

$$\mathrm{Cov}\left(u_i, d_i^*(\kappa)\right) = \mathrm{Cov}(u_i, \epsilon_i) = 0.3. \tag{48}$$

Together with Equation (46), this implies that the jointly normal pair $(u_i, d_i^*(\kappa))$ has the same means, variances, and covariance for every κ. Its joint distribution is therefore unchanged across the relevance designs. Because treatment is a fixed transformation of the latent index, the joint distribution of u_i and $D_i(\kappa)$ is also preserved.

The extension therefore changes the contribution of the excluded z components to treatment while keeping the marginal treatment distribution, the structural outcome equation, the structural quantile treatment effects, the control distribution, the observed-instrument construction, the number of moment conditions, and the estimator and inference procedures fixed. Conditional relationships involving treatment and the observed instruments necessarily change because changing that relationship is the purpose of the experiment.

4.3 Simulation Design and Estimator Comparison

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} &n \in \{500, 1000\}, && \tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ &\kappa \in \{1, 0.50, 0.25, 0.10\}, && R = 500. \end{aligned} \tag{49}$$

Here n is the Monte Carlo sample size, τ is the quantile index, κ is the instrument-relevance design parameter, and $R = 500$ is the number of Monte Carlo replications in each (n, τ, κ) design. The Cartesian product contains 40 distinct (n, τ, κ) designs, each with 500 replications.

Within each Monte Carlo replication, the same primitive draws of u_i, ϵ_i, X_i, z_{1i}, z_{2i}, v_{1i}, v_{2i}, and w_i are reused across all four κ values. This common-random-number construction improves comparability across the relevance settings because differences across κ are evaluated on matched simulated samples. It does not eliminate Monte Carlo uncertainty.

Oracle-GMM uses the known oracle control set; Full-GMM uses all controls without regularization; and DML-IVQR uses regularized high-dimensional nuisances and an orthogonalization-based score construction, with the implementation-specific residualization constant term described in Section 3.3. The design therefore has 120 design–estimator cells. Every comparison uses the truth $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$ from Equation (42).

All estimators use the same sample within a replication. DML-IVQR uses the BC pivotal penalty and same-sample procedure in Section 3.3. Inference is conducted at 95%, and power uses the alternatives defined below.

4.4 Performance Measures and Numerical Inference

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\widehat{\alpha}_r(\tau)$ denote the point estimate in replication r, and let $\alpha_0(\tau)$ be the true value from Equation (42). Replications are indexed by $r = 1, \ldots, R$. For simplicity, we omit the estimator superscript and summarize point-estimation performance using

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R} \sum_{r=1}^{R} \left[\alpha_0(\tau) - \widehat{\alpha}_r(\tau) \right], \\ \text{MAE}(\tau) &= \frac{1}{R} \sum_{r=1}^{R} \left| \widehat{\alpha}_r(\tau) - \alpha_0(\tau) \right|, \\ \text{RMSE}(\tau) &= \left\{ \frac{1}{R} \sum_{r=1}^{R} \left[\widehat{\alpha}_r(\tau) - \alpha_0(\tau) \right]^2 \right\}^{1/2}. \end{aligned} \tag{50}$$

Measures are computed separately by estimator and design cell. Positive Bias means estimates lie below the truth on average; MAE records absolute error, and RMSE gives greater weight to large errors.

In the Monte Carlo experiment, the additional subscript r identifies the replication. With $c_{0.95}$ denoting the critical value defined below, define

$$C_r(\tau) = \mathbf{1}\{W_{n,r}(\alpha_0(\tau)) \le c_{0.95}\}, \qquad \widehat{\text{Coverage}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} C_r(\tau). \tag{51}$$

Coverage evaluates $W_{n,r}$ directly at $\alpha_0(\tau)$ and is therefore independent of the numerical grid. For DML, the pivotal penalty generates fresh Uniform draws on each evaluation, so a direct truth evaluation and a stored grid-profile evaluation at the same numerical candidate need not use identical draws. So, reported DML coverage is the repeated-sample acceptance frequency of the direct candidate-specific procedure. The empirical size is

$$\widehat{\text{Size}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\{W_{n,r}(\alpha_0(\tau)) > c_{0.95}\} = 1 - \widehat{\text{Coverage}}(\tau). \tag{52}$$

The equality follows from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta, \qquad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \tag{53}$$

For each offset, empirical power is

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\{W_{n,r}(a_{\Delta}(\tau)) > c_{0.95}\}. \tag{54}$$

The statistic is evaluated directly at each false value, independently of the confidence-region grid; $\Delta = 0$ is size, not power.

The numerical confidence-set calculation proceeds by test inversion (Chernozhukov and Hansen 2008). For fixed τ and estimator, take a candidate structural effect a, re-estimate the required nuisance quantities conditional on it, construct the estimator-specific IVQR score, and compute $W_n(a)$. Under the null that a is the true effect, the asymptotic reference distribution has two degrees of freedom because there are two instrument moment components. Collecting every candidate that passes its own test gives the confidence region. It is not chosen around the point estimate: minimizing $W_n(a)$ and retaining candidates below a critical value are distinct operations.

Conventional Wald inference centered on an estimate typically relies more directly on regular local identification. Direct testing of each hypothesized effect remains useful under weak or partial identification. This robustness concerns test validity, not narrow regions: many competing effects may survive under weak relevance.

The accepted-set measure totals all accepted components inside the primary computational domain, including disconnected components. It is neither the span between extreme accepted values nor an unrestricted confidence-region length. Because $d_z = 2$, the 95% cutoff for the statistic in Section 3.4 is

$$c_{0.95} = \chi^2_{2,0.95} = 5.991464547\ldots. \tag{55}$$

A candidate is accepted when $W_n(a) \le c_{0.95}$ and rejected when $W_n(a) > c_{0.95}$, with the 95% cutoff 5.991464547 . . . above. The primary grid is

$$\begin{aligned} A_0 &= [-1, 3], & h &= 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \ldots, 41\}, \\ &= \{-1, -0.9, \ldots, 2.9, 3\}. \end{aligned} \tag{56}$$

Here $A_0 = [-1, 3]$ is the fixed primary computational domain, $h = 0.1$ is the distance between adjacent grid points, j is the grid-point index, and a_j denotes the j-th candidate treatment-effect value. The set G_0 contains all 41 candidate values. There are 41 points because the four-unit interval from -1 to 3, evaluated with a step size of 0.1 and including both endpoints, contains $4/0.1 + 1 = 41$ grid points. The original authors' main Monte Carlo implementation uses `seq(-1,3,length=41)`. Preserving its four-unit domain and 0.10 resolution avoids choosing a new grid after observing the extension results. All five quantiles, strength settings, and estimators use this common grid, making their measures comparable without differences in grid density. The resolution is inherited, not claimed to be asymptotically optimal. All five true effects lie strictly inside the domain: the extreme values are $\alpha_0(0.10) \approx -0.282$ and $\alpha_0(0.90) \approx 2.282$, with $\alpha_0(0.50) = 1$.

The distinction between the theoretical region and its finite-grid implementation is important. The theoretical region $C = \{a : W_n(a) \le c_{0.95}\}$ ranges over the relevant parameter space. Computation evaluates only finitely many candidates, producing an accepted set inside A_0, a computational search region rather than an asserted parameter space. The reported summary is therefore the grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, rather than unrestricted confidence-region length.

Total confidence-region length is an intuitive informativeness summary, but numerical measurement requires localization inside a sufficiently wide search domain. The initial design considered adaptive expansion; frozen diagnostics showed that weak-relevance accepted sets could keep reaching wider boundaries. Stopping at an arbitrary endpoint and reporting that finite span would make length partly an endpoint artifact. The main comparison therefore uses fixed A_0, while separate $A_1 = [-3, 5]$ and $A_2 = [-5, 7]$ diagnostics examine endpoint sensitivity (see Appendix A).

For replication r, define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\{W_{n,r}(a_j) \le c_{0.95}\}, \\ C_{r,G_0} &= \{a_j \in G_0 : I_r(a_j) = 1\}, \\ F_r &= \mathbf{1}\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\}. \end{aligned} \tag{57}$$

For the primary domain A_0, define left-boundary, right-boundary, and either-boundary acceptance in replication r by

$$B_r^L = I_r(a_1), \qquad B_r^R = I_r(a_{41}), \qquad B_r^E = \mathbf{1}\{B_r^L = 1 \text{ or } B_r^R = 1\}. \tag{58}$$

The corresponding either-boundary acceptance rate over R Monte Carlo replications is

$$\widehat{B}^E = \frac{1}{R}\sum_{r=1}^{R} B_r^E. \tag{59}$$

The same definition is used for an expanded diagnostic domain, with its own left and right endpoints. A boundary is accepted when the candidate-specific test does not reject the endpoint of the finite numerical search interval. A high boundary-acceptance rate therefore means that accepted components frequently reach the edge of the examined domain and are not localized inside it. An accepted endpoint suggests that a component may continue outside the domain; a measure near four indicates limited localization within A_0. The A_1 and A_2 diagnostics examine whether this behavior disappears. Failure to localize even within A_2 does not establish global unboundedness or failed identification. If boundary acceptance disappears after the domain is widened, the observed accepted grid components no longer touch the investigated endpoints. This does not rule out additional disconnected accepted components outside the investigated domain. Persistent boundary acceptance as successively wider finite domains are examined indicates continued lack of numerical localization over those investigated ranges.

The frozen implementation computes the grid-based accepted-set measure within A_0 as integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2}\left[I_r(a_j) + I_r(a_{j+1})\right]. \tag{60}$$

The sum runs from $j = 1$ to 40 because the 41 grid points in G_0 define 40 adjacent grid intervals $[a_j, a_{j+1}]$. The sum includes separated accepted components without filling rejected gaps and does not interpolate critical-value crossings. With $h = 0.10$, an adjacent interval contributes zero if both endpoints are rejected, 0.10 if both are accepted, and 0.05 if exactly one is accepted. Thus the measure ranges from zero to four and summarizes numerical acceptance, including disconnected components, without solving for exact continuous crossings of $W_n(a)$

with the cutoff. The Monte Carlo mean of F_r is the frequency with which all tested grid points are accepted, not acceptance of the real line.

Coverage evaluates the exact $\alpha_0(\tau)$, whether or not it is a grid point; power evaluates the exact $\alpha_0(\tau)+\Delta$, independently of G_0. Grid and domain choices affect the accepted-set measure and profile point estimate, but do not mechanically determine reported coverage or power. A post-main diagnostic examines sensitivity to the finite endpoints using

$$A_0 = [-1, 3], \qquad A_1 = [-3, 5], \qquad A_2 = [-5, 7]. \tag{61}$$

It uses $n = 1000$, saved primitive samples, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and spacing 0.1. Stage 1 evaluates A_1 and its nested A_0 profile; Stage 2 reloads the same samples and adds only the tails needed for A_2. Domain comparisons therefore change candidates, not samples. The outputs record accepted measure, grid share and count, boundary acceptance, all-grid acceptance, and incremental measure.

The boundary-acceptance diagnostic follows the finite-domain interpretation of Equations (58)–(59). The bounded diagnostic ends at A_2, while main summaries retain A_0 and direct coverage and power remain unchanged.

5 Results and Discussion

The analysis proceeds in six steps: verifying the relevance ordering induced by κ; examining point-estimation accuracy; assessing finite-sample coverage; evaluating power against false candidate values; examining accepted-set informativeness; and drawing the overall interpretation while stating its limits. The central question is how quickly deterioration appears, and through which dimensions, in this high-dimensional IVQR design. Reading these measures together distinguishes inaccurate estimation, poorly calibrated inference, and tests that retain too many competing effects to be informative.

5.1 Instrument-Relevance Calibration

Table 1 checks whether the four values of κ produce the intended ordering of excluded-instrument relevance. In each calibration replication, the restricted OLS regression uses the continuous treatment D as the dependent variable and includes an intercept and $X_1, \ldots, X_{10}$. The unrestricted regression adds Z_1 and Z_2, which are tested jointly. The reported nested-regression F-statistic and partial R^2 are calculated over 100 replications for every (n, κ) de-

sign. They are descriptive first-stage diagnostics rather than formal IVQR identification statistics.

Table 1. Instrument-strength calibration

n	κ	**Median F**	**F: p10–p90**	**Median partial R^2**	**Partial R^2: p10–p90**
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo instrument-relevance diagnostics. The reported dispersion ranges are the 10th and 90th percentiles.

Both diagnostics decrease at every step as κ falls. For $n = 500$, the median F-statistic declines from 101.17 at $\kappa = 1$ to 18.71, 4.74, and 1.22 at the three weaker settings. For $n = 1000$, the corresponding sequence is 212.82, 38.65, 9.85, and 2.04. The dispersion ranges in Table 1 shift in the same direction. Median partial R^2 falls from 0.294 to 0.005 for $n = 500$ and from 0.301 to 0.004 for $n = 1000$, with intermediate values near 0.07 at $\kappa = 0.50$ and 0.02 at $\kappa = 0.25$.

At a fixed κ, partial R^2 is similar across the two sample sizes because it records the incremental explanatory share of the two instruments. The F-statistic nevertheless rises with n because that contribution is estimated more precisely with more observations. The joint movement of the two diagnostics therefore confirms the designed relevance ordering: $\kappa = 1$ provides the strongest relationship and $\kappa = 0.10$ the weakest. This ordering is used to organise the results below; it does not impose a conventional first-stage threshold or establish a universal IVQR identification classification.

5.2 Point-Estimation Accuracy

Table 2 displays finite-sample performance for $n = 1000$, the representative quantiles $\tau \in \{0.10, 0.50, 0.90\}$, all four κ values, and the three estimators. The comparisons in this subsection also use the complete saved results for both sample sizes and all five quantiles, retained in the repository's reporting outputs. Each point estimate is the first minimiser of the

candidate-specific statistic on the ordered 41-point grid in $A_0 = [-1, 3]$, rather than a continuously optimised estimate.

Point accuracy deteriorates systematically as relevance weakens. Across the complete results, both MAE and RMSE increase strictly at every step from $\kappa = 1$ to 0.10 in all 30 estimator–sample-size–quantile sequences. For example, at $n = 1000$ and $\tau = 0.50$, Oracle RMSE rises from 0.072 through 0.134 and 0.431 to 0.945. The corresponding strong-to-weak changes are 0.100 to 1.018 for Full-GMM and 0.104 to 0.991 for DML-IVQR. Deterioration of Oracle-GMM, which knows the designated relevant control set, shows that high-dimensional nuisance estimation is not the sole source of the loss of accuracy. It does not identify which other feature of the design or implementation produces the magnitude of that loss.

Increasing the sample size improves RMSE in every one of the 60 matched estimator–quantile–κ comparisons. At $\tau = 0.50$, Oracle RMSE falls from 0.102 to 0.072 when n rises from 500 to 1000 at $\kappa = 1$, and from 1.027 to 0.945 at $\kappa = 0.10$. At $\tau = 0.90$ and $\kappa = 0.50$, DML RMSE falls from 0.539 to 0.396. These reductions establish better point accuracy in the larger simulated sample, but the remaining errors can still be substantial under weak relevance. Two sample sizes do not establish an asymptotic rate or imply that inference is well calibrated or informative in every design.

Estimator rankings are systematic but not uniform. Across the 40 matched (n, τ, κ) designs, DML has lower RMSE than Full-GMM in 35 and lower RMSE than Oracle-GMM in 11. A sizeable DML advantage occurs at $n = 500$, $\tau = 0.90$, and $\kappa = 0.50$, where RMSE is 0.539 for DML, 1.362 for Full, and 0.822 for Oracle. One clear exception is $n = 500$, $\tau = 0.10$, and $\kappa = 0.50$, where DML RMSE is 1.335 compared with 1.219 for Full. Other reversals are small: at $n = 1000$, $\tau = 0.50$, and $\kappa = 1$, the values are 0.104 and 0.100. These counts are descriptive, and such small differences do not establish statistically meaningful superiority. Oracle is a benchmark with knowledge of the relevant controls, not a guaranteed finite-sample winner; moreover, the estimators differ in residualisation and other implementation details as well as regularisation.

Quantile patterns also distinguish directional bias from overall error. Under the convention Bias $= \alpha_0 - \widehat{\alpha}$, negative bias means overestimation and positive bias means underestimation. Across both sample sizes, all estimators, and all four κ values, the 24 biases at $\tau = 0.10$ are negative, whereas the 24 biases at $\tau = 0.90$ are positive. The estimates therefore move inward on average at the two tail quantiles. No equally uniform sign statement holds for the three interior quantiles. Using unrounded values, the median has lower MAE and lower RMSE than each tail in every fixed-estimator, fixed-n, fixed-κ comparison. This uniform centre-versus-

Table 2. Main finite-sample performance at $n = 1000$

τ	κ	Estimator	Bias	MAE	RMSE	Coverage	Median accepted-set measure
0.10	1.00	Oracle-GMM	-0.054	0.093	0.121	0.860	0.400
		Full-GMM	-0.329	0.337	0.387	0.502	0.800
		DML-IVQR	-0.005	0.117	0.150	0.968	0.600
	0.50	Oracle-GMM	-0.101	0.194	0.287	0.926	1.000
		Full-GMM	-0.407	0.465	0.630	0.804	1.900
		DML-IVQR	-0.159	0.306	0.482	0.948	2.225
	0.25	Oracle-GMM	-0.384	0.566	0.934	0.930	3.450
		Full-GMM	-0.773	0.914	1.242	0.932	3.500
		DML-IVQR	-0.681	0.844	1.229	0.950	3.750
	0.10	Oracle-GMM	-0.747	0.975	1.376	0.946	4.000
		Full-GMM	-1.074	1.251	1.582	0.942	3.900
		DML-IVQR	-0.932	1.146	1.514	0.936	4.000
0.50	1.00	Oracle-GMM	0.007	0.047	0.072	0.872	0.200
		Full-GMM	0.007	0.072	0.100	0.896	0.400
		DML-IVQR	0.070	0.079	0.104	0.776	0.300
	0.50	Oracle-GMM	0.015	0.093	0.134	0.936	0.500
		Full-GMM	0.014	0.151	0.199	0.946	0.800
		DML-IVQR	0.059	0.119	0.160	0.918	0.600
	0.25	Oracle-GMM	0.045	0.281	0.431	0.946	1.900
		Full-GMM	0.029	0.372	0.500	0.936	2.200
		DML-IVQR	-0.003	0.306	0.448	0.948	2.025
	0.10	Oracle-GMM	-0.044	0.735	0.945	0.942	3.900
		Full-GMM	-0.012	0.840	1.018	0.948	3.800
		DML-IVQR	-0.195	0.779	0.991	0.942	3.900
0.90	1.00	Oracle-GMM	0.072	0.110	0.148	0.904	0.500
		Full-GMM	0.387	0.397	0.465	0.530	1.000
		DML-IVQR	0.183	0.186	0.215	0.498	0.400
	0.50	Oracle-GMM	0.211	0.295	0.540	0.940	1.575
		Full-GMM	0.584	0.629	0.871	0.834	2.450
		DML-IVQR	0.226	0.260	0.396	0.844	1.100
	0.25	Oracle-GMM	0.491	0.658	1.038	0.952	3.650
		Full-GMM	0.901	1.058	1.389	0.920	3.600
		DML-IVQR	0.417	0.580	0.927	0.922	3.350
	0.10	Oracle-GMM	0.653	0.948	1.350	0.964	4.000
		Full-GMM	1.050	1.236	1.583	0.948	3.900
		DML-IVQR	0.519	0.831	1.191	0.940	4.000

Note: Each cell is based on 500 Monte Carlo replications. Bias follows the frozen convention (Bias $= \alpha_0 - \widehat{\alpha}$). Coverage is evaluated directly at the true structural quantile effect. The accepted-set measure is the grid-based measure within the prespecified primary computational domain ($A_0 = [-1, 3]$); it is not an unrestricted confidence-region length.

tail ordering in point accuracy does not extend to coverage, which varies differently across quantiles and designs. Mean bias can also conceal large replication errors: for Oracle at $n = 1000$, $\tau = 0.50$, and $\kappa = 0.10$, Bias is only -0.044 while RMSE is 0.945. Positive and negative errors partly cancel in the mean, so small Bias is not evidence of small MAE or RMSE.

The fixed grid affects the point estimator through its resolution, endpoints, and first-tie rule. Yet no diagnostic isolates those constraints or regularisation as the cause of the bias pattern. Overall, accuracy falls as relevance weakens, despite qualified gains from a larger sample and from DML relative to Full-GMM.

5.3 Coverage and Finite-Sample Calibration

Figure 1 shows that empirical coverage differs considerably from the nominal 95% level, even under the strongest instrument relevance considered. At $n = 1000$ and $\kappa = 1$, Oracle coverage is 0.872 at the median, while Full-GMM coverage is only 0.502 and 0.530 at $\tau = 0.10$ and 0.90. DML displays marked tail asymmetry, with coverage 0.968 at $\tau = 0.10$ and 0.498 at $\tau = 0.90$. Figure 1 reports the complete estimator–quantile pattern. Because coverage is evaluated directly at $\alpha_0(\tau)$, these discrepancies are not generated by rounding truth to the numerical grid.

Increasing the sample size can improve point accuracy without improving calibration. At $\kappa = 1$, DML coverage falls from 0.976 to 0.894 between $n = 500$ and $n = 1000$ at $\tau = 0.25$, from 0.922 to 0.776 at $\tau = 0.50$, from 0.770 to 0.632 at $\tau = 0.75$, and from 0.658 to 0.498 at $\tau = 0.90$. These changes equal -8.2, -14.6, -13.8, and -16.0 percentage points. Section 5.2 found lower RMSE at $n = 1000$ in all 60 matched comparisons, so lower observed RMSE of the grid minimiser does not imply that the candidate-specific test has size closer to its nominal level.

Coverage tends to move toward 0.95 as relevance weakens, but not monotonically in every cell. Averaged over estimators, quantiles, and sample sizes, coverage rises from 0.798 at $\kappa = 1$ to 0.912, 0.941, and 0.944 as κ falls. Only 15 of the 30 estimator–sample-size–quantile sequences are nondecreasing at every step. For example, DML coverage at $n = 1000$ and $\tau = 0.10$ is 0.968, 0.948, 0.950, and 0.936 across declining κ, while Full-GMM coverage at $n = 500$ and $\tau = 0.25$ is 0.788, 0.920, 0.938, and 0.930. These reversals are generally small compared with the strong-relevance distortions. More importantly, movement toward nominal coverage under weak relevance can result from rejecting candidate values less often overall.

Each rate is based on $R = 500$ replications. Its Monte Carlo standard error is

$$\mathrm{MCSE}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{R}}.$$

The maximum MCSE is approximately 0.022 and the MCSE at 0.95 is approximately 0.010. Appendix C strengthens this cell-by-cell uncertainty check. Across 60,000 valid direct-at-truth evaluations, 53 pointwise two-sided 95% Wilson intervals lie wholly below 0.95, one lies wholly above it, and 66 contain it. No truth statistic is missing or nonfinite. The intervals use integer acceptance counts without a continuity correction; they are pointwise Monte Carlo intervals, not simultaneous or multiplicity-adjusted tests. An individual rate's MCSE is not used as the standard error of a difference between correlated simulation results.

The audit rules out missing-value handling, nonfinite truth statistics, and reconstruction from rounded grid confidence sets as explanations for the observed undercoverage. It does not isolate a cause among finite-sample approximation error, features of the preserved implementation, or failure of the idealised regularity conditions for the asymptotic chi-square reference law in this DGP. Strong-relevance undercoverage also shows that weak relevance alone cannot

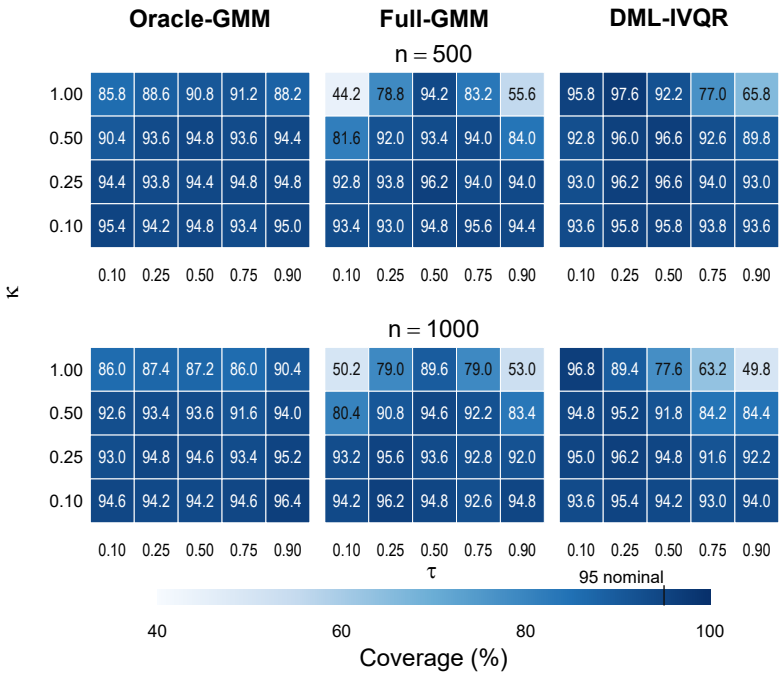


Figure 1. Empirical coverage across simulation designs.

Note: Nominal coverage is 95%; each cell uses 500 complete replications. Coverage is the fraction of tests accepting the true parameter, evaluated directly at that parameter rather than reconstructed from grid membership. Panel columns identify estimators; panel rows show $n = 500$ and $n = 1000$. Within each panel, columns show quantiles τ and rows show relevance κ, declining from 1.00 to 0.10. Annotations and the common colour scale report percentages. Darker shading indicates higher coverage, not necessarily better calibration.

explain the distortions. Coverage records retention of the truth; it does not record how readily false candidates are excluded. The next two sections therefore examine power and accepted-set informativeness before interpreting near-nominal coverage under weak relevance.

5.4 Power Against False Candidate Values

Figure 2 reports rejection probabilities at $\tau = 0.50$ for candidate values $a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta$. At $\Delta = 0$, the candidate is true and rejection probability is empirical size; at each of the six nonzero deviations, it is empirical power against that specified false candidate. The data-generating parameter remains fixed throughout. The plotted probabilities use the same nominal 5% chi-square cutoff and are not adjusted for differences in empirical size.

At $n = 1000$ and $\Delta = -0.50$, Oracle power falls from 1.000 through 0.850 and 0.316 to 0.082 as κ declines, while DML falls from 1.000 through 0.802 and 0.312 to 0.104. Full-GMM likewise falls from 0.998 to 0.076 between the strongest and weakest settings. For $\Delta = 0.50$ at $\kappa = 0.25$, power is 0.378, 0.238, and 0.286 for Oracle, Full, and DML, before falling to approximately one tenth for all three at $\kappa = 0.10$. The deterioration is therefore already substantial at intermediate relevance, and false effects separated from truth by one half become difficult to exclude in the weakest setting.

Increasing n usually raises power where usable signal remains. At $\kappa = 0.50$, Oracle power at $\Delta = -0.50$ rises from 0.552 to 0.850, Full from 0.260 to 0.634, and DML from 0.466 to 0.802 between $n = 500$ and $n = 1000$. At $\Delta = 0.50$, the corresponding changes are 0.560 to 0.940, 0.276 to 0.726, and 0.392 to 0.852. At $\kappa = 0.10$, improvements are much smaller and not uniform: Full power at $\Delta = -0.50$ falls from 0.088 to 0.076, while DML remains 0.104; at $\Delta = 0.50$, DML rises from 0.030 to 0.094. Across all five quantiles and six nonzero deviations, $n = 1000$ has higher rejection probability in 305 of 360 matched cells, the same probability in 25, and a lower probability in 30.

The null points clarify why raw power cannot be used as an efficiency ranking. At $n = 1000$ and $\kappa = 1$, median empirical size is 0.128 for Oracle, 0.104 for Full, and 0.224 for DML, compared with 0.092, 0.058, and 0.078 at $n = 500$. DML's strong-relevance rejection of false candidates therefore coexists with the size distortion documented in Section 5.3. At $n = 1000$ and $\kappa = 0.10$, empirical sizes are 0.058, 0.052, and 0.058, while power at $\Delta = -0.50$ is only 0.082, 0.076, and 0.104. Rejection of this false candidate is then close to rejection at truth, indicating little discrimination despite approximately nominal size.

The median pattern broadly generalises, but it is not universally monotone. Across all quantiles, sample sizes, estimators, and nonzero deviations, power is nonincreasing as κ falls in

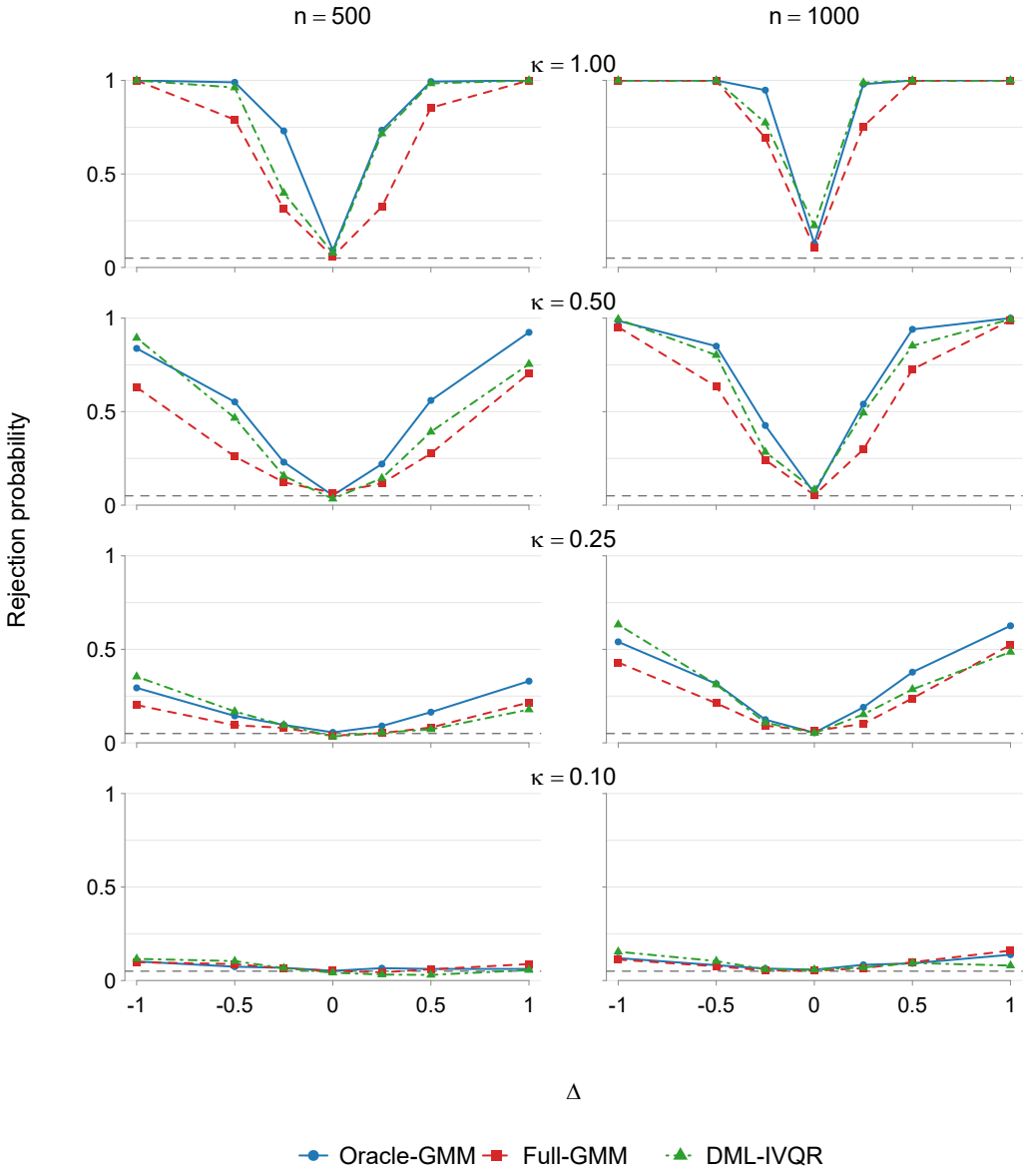


Figure 2. Empirical power at the median.

Note: The median benchmark is $\tau = 0.50$. Columns show $n = 500$ and $n = 1000$; rows show $\kappa = 1.00, 0.50, 0.25, 0.10$. The horizontal axis is $\Delta = a - \alpha_0(0.50)$, the saved deviation of the tested candidate from truth. Each point uses 500 complete replications and the same nominal 5% chi-square cutoff; dashed horizontal lines mark 0.05. Rejection at truth is evaluated directly. At $\Delta = 0$, rejection probability measures empirical size; at nonzero deviations, it measures power. These rejection probabilities are not size-adjusted. Straight segments are visual guides between evaluated points.

162 of 180 sequences. The 18 exceptions are mostly small reversals when rejection is already close to empirical size. Signed differences also occur: for DML at $\tau = 0.50$, $n = 500$, and $\kappa = 0.25$, power is 0.168 at $\Delta = -0.50$ but 0.072 at $\Delta = 0.50$. Figure 2 displays only $\tau = 0.50$, so these full-result counts support generalisation without implying that every quantile follows the plotted paths. The design does not isolate the source of signed asymmetry, and asymmetry in power is not by itself evidence of point-estimation bias.

5.5 Confidence-Set Informativeness

Figure 3 reports the median grid-based accepted-set measure within $A_0 = [-1, 3]$. The measure increases strictly at every step as κ falls in all 30 estimator–sample-size–quantile sequences. At $n = 1000$ and $\tau = 0.50$, Oracle, Full, and DML medians rise from 0.200, 0.400, and 0.300 at $\kappa = 1$ to 3.900, 3.800, and 3.900 at $\kappa = 0.10$. Thus declining relevance causes progressively more of the maintained domain to survive test inversion, even when coverage moves closer to 0.95.

Quantile and estimator differences are visible before the weakest setting. At $n = 1000$ and $\kappa = 0.50$, DML's median measure ranges from 0.600 at $\tau = 0.50$ to 2.225 at $\tau = 0.10$, while Full ranges from 0.800 to 2.450 across the five quantiles. Increasing the sample size narrows the median accepted measure in 52 of 60 matched cells and leaves it unchanged in eight; it never widens it. For example, DML at $\tau = 0.50$ and $\kappa = 0.25$ falls from 3.050 to 2.025, whereas at $\kappa = 0.10$ it falls only from 4.000 to 3.900. More observations improve localisation, but do not overcome the weakest relevance setting at these sample sizes.

The DML sequence at $n = 1000$ and $\tau = 0.50$ connects the four inferential summaries. At $\kappa = 1$, coverage is 0.776, the median accepted measure is 0.300, the full-grid acceptance rate is 0.000, and power at $\Delta = 0.50$ is 1.000. At $\kappa = 0.25$, the corresponding values are 0.948, 2.025, 0.040, and 0.286. At $\kappa = 0.10$, coverage remains near nominal at 0.942, but the median measure is 3.900, full-grid acceptance is 0.428, and power at $\Delta = 0.50$ is 0.094. Truth retention, the amount of the grid retained, acceptance of the entire grid, and rejection of a particular false value therefore answer different questions.

The measure cannot exceed four because it integrates grid acceptance only over $A_0 = [-1, 3]$. Similar values near four do not imply equivalent unrestricted confidence regions: the calculation does not examine candidates outside A_0, and distinct acceptance patterns can produce similar totals. Appendix Figure C.1 records the frequency with which all 41 evaluated points are accepted. That event does not establish acceptance at every intermediate real value or anywhere outside A_0. Coverage is also a separate direct evaluation at truth. For DML, fresh

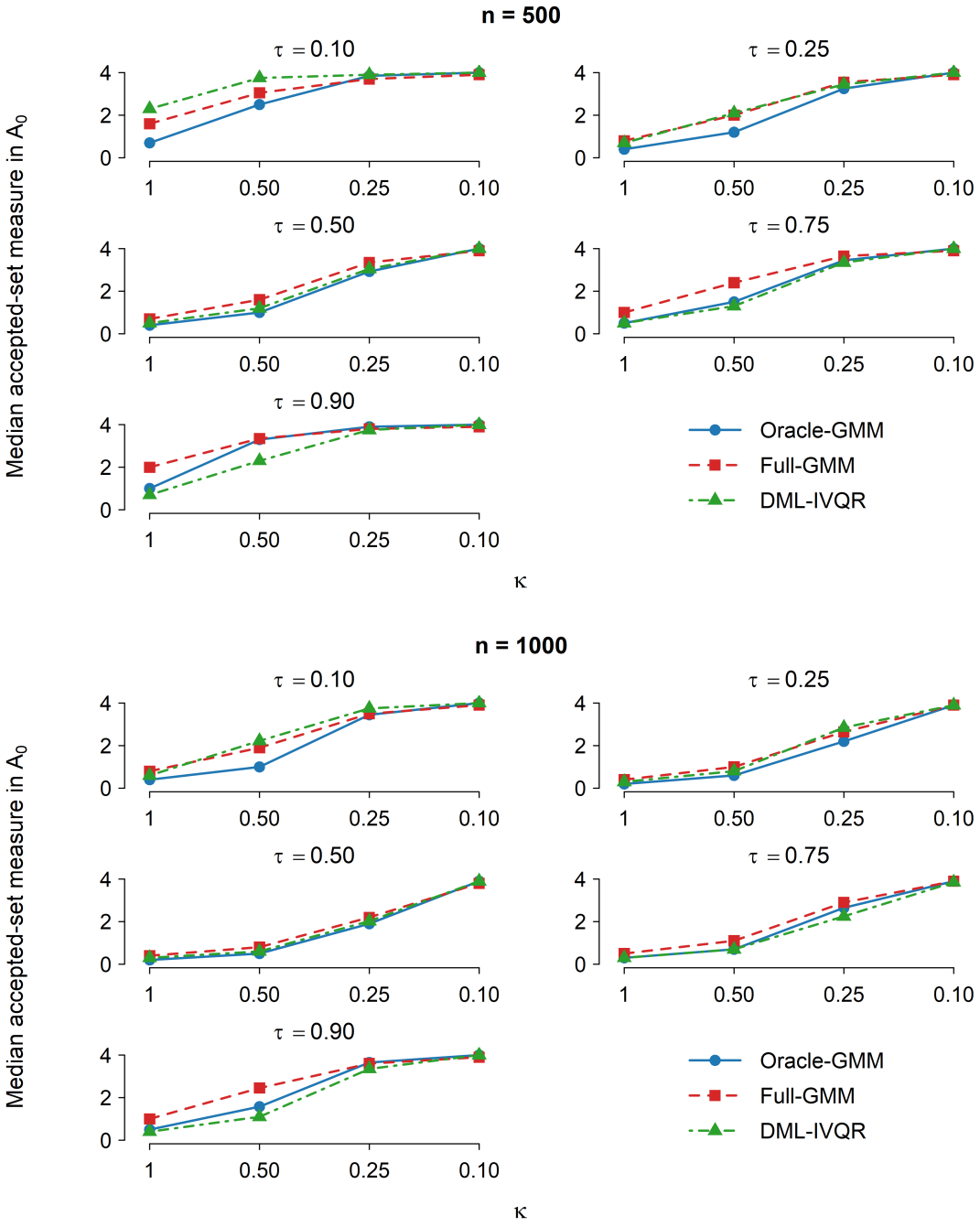


Figure 3. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

pivotal draws mean that direct-truth coverage and stored-profile acceptance need not coincide even when truth is numerically a grid point.

Appendix A examines wider domains in a separate diagnostic with $n = 1000$, 100 replications, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and step 0.1 on $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. At $\tau = 0.50$ and $\kappa = 1$, median DML measure remains 0.300 on all three domains and A_2 boundary acceptance is zero. At $\kappa = 0.25$, it grows from 1.950 to 2.500 and 3.025, with A_2 boundary acceptance 0.500. At $\kappa = 0.10$, it grows from 3.900 to 7.500 and 10.325, and the A_2 boundary is accepted in 0.960 of usable profiles. Oracle and Full display the same broad weak-relevance pattern, with A_2 median measures 11.300 and 11.450 and boundary rates 0.830 and 0.850 in the corresponding cell.

These domain-expansion values come from a diagnostic run with 100 replications, not the main 500-replication experiment. They show near-stability within the investigated domains under strong relevance and continued failure to localise at intermediate and weak relevance. The diagnostic ends at A_2 and does not search the real line. Failure to localise over these domains is evidence of limited numerical informativeness, not proof of global unboundedness or non-identification.

5.6 Overall Interpretation and Limitations

Taken together, the results show that weakening instrument relevance changes each dimension of finite-sample performance, but not in the same way. Point accuracy deteriorates systematically, accepted sets expand, and power against false candidate values generally declines, subject to the documented small reversals. Coverage is less systematic and can move toward its nominal level precisely when the procedure rejects few candidates. Near-nominal coverage can therefore coexist with low power and acceptance of most or all of the numerical domain. Accuracy, calibration, power, and informativeness measure different features of the procedure, and no one of them is sufficient to assess inference.

The estimator comparison reinforces this interpretation. Oracle-GMM knows the designated relevant controls, yet its point accuracy, power, and accepted-set localisation still deteriorate as relevance declines. High-dimensional nuisance estimation therefore cannot be the sole explanation for the common relevance pattern. DML-IVQR usually has lower RMSE than Full-GMM, but the documented exceptions and the separate coverage and power results rule out uniform superiority. Oracle is likewise an infeasible benchmark rather than a guaranteed finite-sample winner. Because the implementations differ in nuisance dimension, regularisa-

tion, and residualisation details, these are comparisons among complete estimator implementations; they do not isolate the causal effect of regularisation.

Sample size and quantile comparisons require similar qualifications. Increasing n from 500 to 1000 lowers RMSE in all 60 matched comparisons and typically narrows accepted sets and raises power where usable signal remains. It does not improve every inferential dimension: several strong-relevance DML coverage rates deteriorate substantially, and power has documented exceptions. Across quantiles, the lower-tail biases are uniformly negative and the upper-tail biases uniformly positive under the convention Bias $= \alpha_0 - \widehat{\alpha}$. At fixed estimator, n, and κ, both MAE and RMSE are lower at the median than at either tail throughout the final results, whereas coverage has no corresponding universal centre-versus-tail ordering. Results at two sample sizes establish finite-sample contrasts, not an asymptotic rate or limit.

Neyman orthogonality is designed to reduce first-order sensitivity to small nuisance-estimation errors (Chernozhukov et al. 2018), while candidate-specific test inversion is designed to avoid relying on a precisely estimated structural coefficient under weak identification (Chernozhukov and Hansen 2008). Neither property creates identifying variation. Subject to its assumptions and regularity conditions, weak-identification-robust construction concerns validity of testing a candidate; it does not promise a narrow confidence region or high power when many false candidates are observationally difficult to distinguish. The DML-IVQR framework of Chen et al. (2021) combines high-dimensional nuisance estimation with this style of candidate-specific inference, but the present simulations evaluate the preserved executable procedure. They do not validate the ideal asymptotic theory.

The population density weight at the true parameter encounters the non-finite regularity problem derived in Section 3.2. The executable estimators instead use the fitted-normal density proxy and implementation-specific residualisation rules in Section 3.3; Appendix D further records that the preserved `dnorm` call supplies the residual variance in the argument position interpreted by R as a standard deviation. DML uses the same sample for nuisances and scores rather than cross-fitting, and its pivotal penalty generates fresh draws across evaluations. All three estimators use an asymptotic chi-square reference cutoff, while their empirical sizes differ. These features qualify the relationship between the experiment and the theoretical procedures, but the design does not vary them separately. None can therefore be identified as the cause of a particular distortion, and the results do not establish that changing one feature would remove it.

The numerical and design scope imposes further limits. Point estimates minimize a 41-point profile and hence reflect its resolution, endpoints, and first-tie rule. The accepted-set measure is restricted to $A_0 = [-1, 3]$ and cannot describe acceptance outside that domain;

Appendix A shows that accepted measure and boundary acceptance can continue to grow on wider investigated domains under weak relevance. Those diagnostics establish limited numerical localisation over the ranges examined, not global unboundedness or non-identification. More broadly, the evidence comes from one DGP family, five quantiles, two sample sizes, four relevance settings, and three preserved estimator implementations. It cannot establish how every alternative DGP, implementation, or larger sample would behave.

The bounded implication is therefore to distinguish what can be assessed in a simulation from what can be established in an empirical application. A procedure can produce a more accurate point estimate while remaining poorly calibrated, attain coverage near the nominal level while excluding few false candidates, or show a finite-domain measure near its ceiling without establishing the shape of an unrestricted confidence region. Simulation evaluations should report accuracy, calibration, power, and informativeness jointly. In empirical applications, researchers should disclose numerical-domain sensitivity and the assumptions underlying inference; nominal confidence levels alone do not establish finite-sample reliability. Within the scope of this experiment, that distinction gives the clearest account of what is lost as instrument relevance weakens.

6 Conclusion

In our thesis we examined how weakening excluded-instrument relevance changes the finite-sample performance and informativeness of the preserved DML-IVQR procedure, and whether those changes differ across structural quantiles. Its contribution is a controlled Monte Carlo extension of the framework in Chen et al. (2021), rather than a new estimator or theory of inference. The design reduces the excluded signal through the parameter κ while adding compensating noise to preserve the marginal treatment distribution and the relationship that generates endogeneity. It then compares Oracle-GMM, Full-GMM, and DML-IVQR across quantiles and sample sizes.

Point-estimation errors increase, numerical accepted sets expand, and tests generally reject false candidate effects less often. Coverage behaves differently: it can move closer to its nominal level while the test becomes less able to distinguish the truth from alternatives. The contrast is especially clear for DML-IVQR at $n = 1000$, $\tau = 0.50$, and $\kappa = 0.10$. Coverage is 0.942, yet the median accepted-set measure is 3.900 within a domain of width four, and all 41 grid points are accepted in 0.428 of replications. These are distinct simulation summaries, and their underlying acceptance events need not coincide replication by replication. Together,

however, they show why frequent retention of the truth does not by itself establish informative inference. Calibration concerns rejection at the truth; informativeness concerns the ability to exclude false values.

Oracle-GMM knows the designated relevant controls, but its accuracy, power, and numerical localisation still deteriorate as relevance weakens. High-dimensional nuisance estimation therefore cannot be the sole explanation for the common decline in performance. DML-IVQR often improves point accuracy relative to Full-GMM, showing that regularised nuisance estimation can be useful in this high-dimensional design, but the advantage is neither universal across design cells nor transferable automatically to coverage or raw power. Oracle-GMM is likewise an informative benchmark rather than a guaranteed finite-sample winner. Moreover, the three procedures differ in their complete implementations, including nuisance dimension and residualisation as well as regularisation. Their comparison therefore does not isolate the causal effect of regularisation, and differences in unadjusted power cannot be read as pure efficiency rankings when empirical sizes also differ.

Sample size and quantile comparisons reveal a similar separation among performance measures. Raising n from 500 to 1000 lowers observed RMSE in every matched estimator–quantile–κ comparison, but it does not ensure improved calibration. Several DML-IVQR coverage rates deteriorate substantially in the strong-relevance design even as point accuracy improves. Across quantiles, the final results have a clear accuracy ordering: at fixed estimator, sample size, and κ, median MAE and RMSE are lower than at either $\tau = 0.10$ or $\tau = 0.90$ throughout. Coverage has no corresponding universal centre-versus-tail ordering. The evidence therefore supports finite-sample statements about these two sample sizes; it neither identifies an asymptotic rate nor shows that lower RMSE reflects lower estimator variance alone.

Neyman orthogonality is intended to reduce first-order sensitivity to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). Candidate-specific test construction is intended to accommodate weak identification under its assumptions without relying on a precisely estimated structural coefficient (Chernozhukov and Hansen 2008). Neither property creates identifying variation, guarantees accurate finite-sample critical values, or ensures that false candidate effects can be excluded. The simulations illustrate the practical importance of evaluating nuisance robustness, finite-sample calibration, power, and accepted-set informativeness separately. They do not establish or refute the general asymptotic theory of DML-IVQR; they evaluate the executable procedure preserved for this experiment.

That scope matters because the relationship between the executable procedure and the ideal construction is qualified in several ways. For this DGP, the population density weight at the true parameter encounters the non-finite regularity problem derived in Section 3.2.

The implemented procedure instead uses a fitted-normal density proxy and implementation-specific residualisation rules, as described in Section 3.3. Appendix D records that the preserved `dnorm` call supplies a residual variance in the argument position interpreted by R as a standard deviation. The DML implementation estimates nuisance functions and scores on the same sample rather than using cross-fitting, and fresh pivotal draws introduce algorithmic randomness across candidate evaluations. All three estimators use an asymptotic chi-square cutoff even though their empirical sizes differ. These features may matter for finite-sample behaviour, but the experiment does not vary them one at a time. It therefore cannot attribute any particular distortion to one implementation choice or establish that changing that choice would restore calibration.

Point estimates are first minimisers on a 41-point grid and consequently reflect its resolution, endpoints, and tie rule. The main accepted-set measure describes only the maintained domain $A_0 = [-1, 3]$. The domain-expansion exercise in Appendix A shows that acceptance can continue over wider investigated domains when relevance is weak, but continued acceptance on a finite sequence of grids is not proof of global unboundedness or non-identification. More broadly, the results come from one DGP family, selected relevance settings, two sample sizes, and five quantiles. They document robust patterns within that design but cannot determine how every alternative DGP, implementation, or larger sample would behave.

These limitations point to focused extensions. First, controlled variants could alter the density proxy, residualisation, sample splitting, or treatment of algorithmic randomness one feature at a time while retaining the present implementation as an explicit baseline and applying common evaluation criteria. Such comparisons would assess sensitivity and possible mechanisms without presuming that any single modification must repair the observed calibration. Second, a complementary DGP satisfying the relevant density-weight regularity requirements could help distinguish properties of the present simulation environment from the behaviour of the theoretical procedure under its maintained assumptions. Third, additional sample sizes and relevance settings could be combined with systematic grid and domain checks. Direct-at-truth coverage should remain distinct from numerical inversion in that work: more replications improve Monte Carlo precision but do not remove systematic miscalibration, and no finite grid can establish global boundedness.

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Appendix A Numerical-Domain Diagnostics

The primary computational domain is $A_0 = [-1, 3]$, evaluated on 41 equally spaced candidate values with step size $h = 0.1$. Using the same domain for all sample sizes, quantiles, instrument-relevance settings, and estimators makes the accepted-set measures directly comparable across the simulation design.

The interval also contains the true structural quantile effects for all five quantiles considered in the thesis. Under the data-generating process,

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau),$$

so the true effects are approximately -0.282, 0.326, 1.000, 1.674, and 2.282 for $\tau = 0.10, 0.25, 0.50, 0.75$, and 0.90, respectively. All five structural truths lie strictly inside A_0; using their unrounded values, the minimum distance to either endpoint is approximately 0.718. The interval should therefore be understood as a common numerical search domain, not as an assumed theoretical parameter space.

A finite search domain can nevertheless become restrictive when instruments are weak and many candidate values are accepted. For this reason, a separate sensitivity diagnostic expands the search range from $A_0 = [-1, 3]$ to $A_1 = [-3, 5]$ and then to $A_2 = [-5, 7]$. Under strong instrument relevance, expanding the domain adds little accepted measure, indicating that the observed accepted components are well localized within the investigated ranges. At intermediate relevance, the wider domains contain additional accepted values in some profiles. Under the weakest relevance setting, the accepted measure continues to increase as the domain is expanded and does not stabilize within A_2.

These results clarify the interpretation of the main accepted-set measure. A value close to four in A_0 means that almost the entire four-unit primary domain is accepted; it does not mean that the unrestricted confidence region has length four. Similarly, continued acceptance in the wider domains indicates poor numerical localization over the ranges examined, but it does not by itself establish that the theoretical confidence region is globally unbounded or that identification completely fails.

Table A.1 retains the key frozen domain-measure comparisons; Figure A.1 summarizes the expansion pattern.

Table A.1. Grid-domain diagnostic: accepted-set measures

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	1.00	Oracle-GMM	0.400	0.400	0.400	0.386	0.386	0.386
1000	0.10	1.00	Full-GMM	0.800	0.800	0.800	0.757	0.757	0.757
1000	0.10	1.00	DML-IVQR	0.600	0.600	0.600	0.705	0.705	0.705
1000	0.10	0.25	Oracle-GMM	3.400	5.325	7.250	3.019	4.994	6.838
1000	0.10	0.25	Full-GMM	3.375	5.950	8.200	3.304	5.774	8.037
1000	0.10	0.25	DML-IVQR	3.750	5.950	7.950	3.607	6.040	8.245
1000	0.10	0.10	Oracle-GMM	4.000	7.900	11.900	3.745	7.381	11.002
1000	0.10	0.10	Full-GMM	3.900	7.800	11.600	3.785	7.394	10.975
1000	0.10	0.10	DML-IVQR	4.000	7.575	10.775	3.845	7.219	10.339
1000	0.25	1.00	Oracle-GMM	0.300	0.300	0.300	0.259	0.259	0.259

Continued on next page

TableA.1. Grid-domain diagnostic: accepted-set measures (Continued)

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.25	1.00	Full-GMM	0.400	0.400	0.400	0.429	0.429	0.429
1000	0.25	1.00	DML-IVQR	0.300	0.300	0.300	0.346	0.346	0.346
1000	0.25	0.25	Oracle-GMM	2.175	2.800	2.850	2.224	3.158	3.964
1000	0.25	0.25	Full-GMM	2.450	3.500	4.075	2.517	3.768	4.745
1000	0.25	0.25	DML-IVQR	2.750	4.650	6.225	2.630	4.130	5.712
1000	0.25	0.10	Oracle-GMM	3.900	7.700	11.500	3.492	6.699	9.872
1000	0.25	0.10	Full-GMM	3.825	7.600	11.100	3.595	6.891	10.083
1000	0.25	0.10	DML-IVQR	4.000	7.700	11.125	3.651	6.926	9.984
1000	0.50	1.00	Oracle-GMM	0.200	0.200	0.200	0.235	0.235	0.235
1000	0.50	1.00	Full-GMM	0.400	0.400	0.400	0.373	0.373	0.373
1000	0.50	1.00	DML-IVQR	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.50	0.25	Oracle-GMM	1.875	2.200	2.200	1.960	2.682	3.229
1000	0.50	0.25	Full-GMM	2.250	2.400	2.550	2.368	3.171	3.802
1000	0.50	0.25	DML-IVQR	1.950	2.500	3.025	2.124	3.020	3.942
1000	0.50	0.10	Oracle-GMM	3.825	7.600	11.300	3.521	6.640	9.687
1000	0.50	0.10	Full-GMM	3.900	7.625	11.450	3.536	6.718	9.817
1000	0.50	0.10	DML-IVQR	3.900	7.500	10.325	3.538	6.671	9.585
1000	0.75	1.00	Oracle-GMM	0.300	0.300	0.300	0.290	0.290	0.290
1000	0.75	1.00	Full-GMM	0.500	0.500	0.500	0.512	0.512	0.512
1000	0.75	1.00	DML-IVQR	0.300	0.300	0.300	0.279	0.279	0.279
1000	0.75	0.25	Oracle-GMM	2.825	4.200	4.850	2.628	3.945	4.990
1000	0.75	0.25	Full-GMM	2.950	4.450	5.150	2.827	4.267	5.478
1000	0.75	0.25	DML-IVQR	2.325	3.300	3.900	2.402	3.595	4.788
1000	0.75	0.10	Oracle-GMM	3.975	7.750	11.700	3.670	6.999	10.357
1000	0.75	0.10	Full-GMM	3.900	7.750	11.500	3.661	7.023	10.267
1000	0.75	0.10	DML-IVQR	3.900	7.350	10.375	3.554	6.691	9.736
1000	0.90	1.00	Oracle-GMM	0.500	0.500	0.500	0.519	0.519	0.519
1000	0.90	1.00	Full-GMM	1.000	1.000	1.000	1.026	1.031	1.031
1000	0.90	1.00	DML-IVQR	0.400	0.400	0.400	0.454	0.454	0.454
1000	0.90	0.25	Oracle-GMM	3.650	5.850	7.850	3.381	5.625	7.675
1000	0.90	0.25	Full-GMM	3.675	6.150	8.575	3.555	6.123	8.432
1000	0.90	0.25	DML-IVQR	3.450	5.400	7.300	3.114	5.249	7.374
1000	0.90	0.10	Oracle-GMM	4.000	7.900	11.800	3.712	7.241	10.771
1000	0.90	0.10	Full-GMM	3.900	7.700	11.500	3.817	7.448	11.031
1000	0.90	0.10	DML-IVQR	4.000	7.700	11.600	3.644	7.042	10.395

Note: $n = 1000$; 100 replications per design; $\kappa \in \{1.00, 0.25, 0.10\}$. Domains $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$ use $h = 0.1$ (41, 81, 121 points). Med. and Mean are replication medians and means of trapezoidal accepted-set measures computed from adjacent grid acceptance indicators within each domain. Summaries use nonmissing measures: DML-IVQR at $\tau = 0.90$, $\kappa = 0.10$ has one failed profile, leaving 99 replications in each expanded domain; all other displayed cells use 100. These numerical diagnostics do not establish global boundedness, unboundedness, or disconnected components.

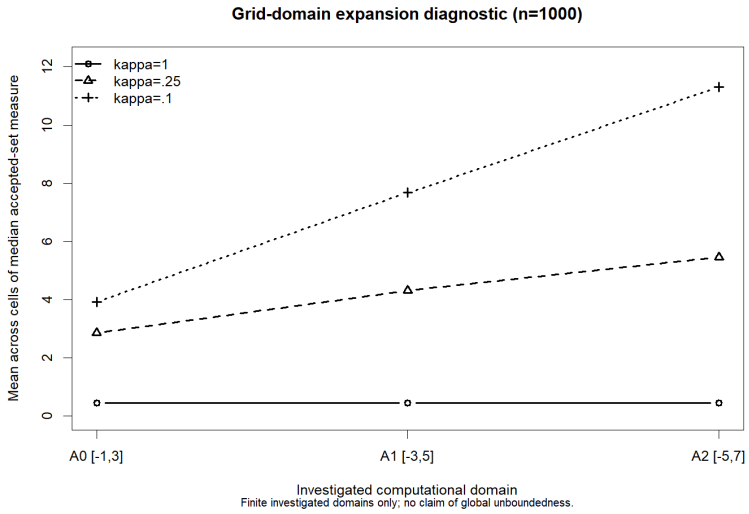


Figure A.1. Sensitivity of the accepted set to expansion of the computational domain

Note: Each plotted value is the mean across 15 estimator–quantile cells of the cell-specific median accepted-set measure for the stated κ and domain. The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

Figure C.1 in Appendix C complements these expansion diagnostics by reporting how often all candidates within the primary grid are accepted.

Appendix B Derivation and Properties of the κ-Extension

B.1 Baseline and Design Objective

This appendix explains how the experiment weakens the instruments' contribution to treatment while preserving the treatment distribution. The baseline assumptions come from the executable design of Chen et al. (2021), described in Section 4.1. The relevance parameter and compensating noise are additions made by this thesis; the coefficient on that noise is derived below.

For observation i, the primitive excluded components z_{1i} and z_{2i} are the parts of the instruments that enter treatment assignment directly. The assignment disturbance ϵ_i also enters treatment, while the outcome disturbance u_i generates unobserved variation in the treatment effect. Their joint distribution is specified in Equation (33): both have mean zero and variance one, with covariance 0.3. This 0.3 is an assumed baseline parameter, not an estimate or a result of the extension. Because both variances equal one, it is also their correlation.

The variables $z_{1i}, z_{2i}, v_{1i}, v_{2i}$ and x_{ji}, $j = 1, \ldots, 100$, are mutually independent standard normals, independent of the pair (u_i, ϵ_i). A standard-normal variable has mean zero and variance one. Observations are independent across i. The observed controls are $X_{ji} = \Phi(x_{ji})$. Here Φ is the standard-normal cumulative distribution function, and Φ^{-1} is its quantile function. The observed instruments Z_{1i}, Z_{2i}

combine the primitive z components, independent noise v, and selected observed controls; their exact equations are given below. Uppercase Z therefore does not denote the same variable as lowercase z.

The parameter κ is a chosen experimental setting, not an estimated quantity. The simulations use $1.00, 0.50, 0.25, 0.10$; the construction is defined for the whole interval $[0, 1]$. We write $d_i^*(\kappa)$ for the latent treatment index, an unobserved real-valued assignment variable, and $D_i(\kappa) = \Phi(d_i^*(\kappa))$ for the observed treatment. At the baseline setting,

$$d_i^*(1) = z_{1i} + z_{2i} + \epsilon_i.$$

To decide what must be preserved, first calculate its variance. For the excluded component, the variance-of-a-sum identity gives

$$\begin{aligned}\operatorname{Var}(z_{1i} + z_{2i}) &= \operatorname{Var}(z_{1i}) + \operatorname{Var}(z_{2i}) + 2\operatorname{Cov}(z_{1i}, z_{2i}) \\ &= 1 + 1 + 0 = 2.\end{aligned}$$

The number two is the combined variance of two independent unit-variance variables. Since ϵ_i is independent of them and has variance one, the complete baseline index has variance $2 + 1 = 3$.

Now consider simply replacing the excluded component by $\kappa(z_{1i} + z_{2i})$. This would give

$$\operatorname{Var}[\kappa(z_{1i} + z_{2i}) + \epsilon_i] = 2\kappa^2 + 1.$$

For example, at $\kappa = 0.50$ the variance would be $2(0.50)^2 + 1 = 1.5$, rather than three. The index remains normal but its distribution changes. Applying Φ then also changes the marginal treatment distribution. Simple scaling therefore changes two things at once: instrument contribution and treatment dispersion.

To isolate the first change, this thesis introduces a new standard-normal variable w_i, independent of all baseline variables. Its purpose is to replace the removed signal variance with independent noise. This is a specific experimental design choice, not an assumption inherited from the source authors.

B.2 Deriving the Compensating Coefficient

The next step is to find the amount of independent noise needed to retain the baseline variance of three. Begin with an unknown nonnegative coefficient $c(\kappa)$ on w_i:

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + c(\kappa)w_i + \epsilon_i, \qquad \kappa \in [0, 1]. \tag{B.1}$$

For a constant a and random variable V, $\operatorname{Var}(aV) = a^2 \operatorname{Var}(V)$. Thus multiplying a variable by a constant scales its variance quadratically. Applying this identity to the excluded component, whose variance is two, and to w_i, whose variance is one, gives

$$\operatorname{Var}[\kappa(z_{1i} + z_{2i})] = 2\kappa^2, \qquad \operatorname{Var}[c(\kappa)w_i] = c(\kappa)^2.$$

The variance of a sum also includes twice each pairwise covariance. Here those terms vanish: $\operatorname{Cov}(z_{1i} + z_{2i}, w_i) = 0$, $\operatorname{Cov}(z_{1i} + z_{2i}, \epsilon_i) = 0$, and $\operatorname{Cov}(w_i, \epsilon_i) = 0$, by the stated independence assumptions. This does not make ϵ_i independent of u_i; their dependence is deliberately retained, and u_i is not a separate component of the index. Adding the three component variances yields

$$\operatorname{Var}\left[d_i^*(\kappa)\right] = 2\kappa^2 + c(\kappa)^2 + 1.$$

We now impose the design requirement that this sum equal the baseline variance of three. Subtracting the disturbance variance of one and the remaining signal variance gives

$$\begin{aligned} 2\kappa^2 + c(\kappa)^2 + 1 &= 3, \\ c(\kappa)^2 &= 3 - 1 - 2\kappa^2, \\ c(\kappa)^2 &= 2 - 2\kappa^2, \\ c(\kappa)^2 &= 2(1 - \kappa^2), \\ c(\kappa) &= \sqrt{2(1 - \kappa^2)}. \end{aligned} \tag{B.2}$$

The term κ^2 appears because variance scales quadratically. The square root is needed because $c(\kappa)$ is a coefficient, not a variance: its square is the variance contributed by unit-variance w_i. We choose the nonnegative root by convention. Symmetry of independent normal noise makes either sign distributionally equivalent. The range $\kappa \in [0, 1]$ ensures that the required added variance is nonnegative. This coefficient follows from the chosen independent-noise construction; it is not a universal weak-instrument formula.

For a worked example, take $\kappa = 0.50$. The remaining signal variance is $2(0.50)^2 = 0.5$. The missing signal variance is $2 - 0.5 = 1.5$, so $c(0.50) = \sqrt{1.5} \approx 1.2247$. The added noise variance is exactly $(\sqrt{1.5})^2 = 1.5$; squaring the displayed rounded coefficient gives approximately the same value. Hence the complete variance is $0.5 + 1.5 + 1 = 3$. Calculations use the unrounded coefficient; rounding is only for display. Independent noise restores dispersion but does not create new instrument signal.

The resulting assignment equation therefore is

$$\begin{aligned} d_i^*(\kappa) &= \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)}\, w_i + \epsilon_i, \\ D_i(\kappa) &= \Phi\big(d_i^*(\kappa)\big)\,. \end{aligned} \tag{B.3}$$

B.3 Preserving the Marginal Treatment Distribution

The variance calculation answers only part of the distributional question. The index is normal because it is a linear combination of jointly normal primitives. Each component has mean zero, so the index has mean zero. Its variance is three for every setting, giving

$$d_i^*(\kappa) \sim \mathcal{N}(0, 3) \qquad \text{for every } \kappa \in [0, 1].$$

In this notation, three denotes the variance, not the standard deviation; the standard deviation is $\sqrt{3}$. Equal variance alone would not establish equal distributions. Here normality means that the unchanged mean and variance determine the entire index distribution. Applying the same function Φ to identically distributed indices therefore produces identically distributed treatments.

To see the treatment distribution explicitly, let d be a possible treatment threshold in $(0, 1)$. Substitute the definition of treatment, then use the fact that Φ is strictly increasing to apply its inverse to both sides of the inequality. Finally, divide the zero-mean index by its standard deviation $\sqrt{3}$ to obtain a standard normal variable:

$$\begin{aligned} \Pr[D_i(\kappa) \le d] &= \Pr\big[\Phi(d_i^*(\kappa)) \le d\big] \\ &= \Pr\big[d_i^*(\kappa) \le \Phi^{-1}(d)\big] \\ &= \Phi\left(\frac{\Phi^{-1}(d)}{\sqrt{3}}\right). \end{aligned} \tag{B.4}$$

Treatment lies in $(0, 1)$, so its CDF is zero for $d \leq 0$ and one for $d \geq 1$. The formula contains no κ, which establishes marginal invariance. It is not the uniform CDF d on $(0, 1)$: applying the standard-normal CDF to an index with variance three does not give a Uniform$(0, 1)$ variable.

B.4 Changing Instrument Relevance

To describe the relationship that changes, consider the observed instruments in Equations (36)–(37):

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i},$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}.$$

Write $C_{1i} = X_{2i} + X_{3i} + X_{4i}$ and $C_{2i} = X_{7i} + X_{8i} + X_{9i} + X_{10i}$. Although these control sums have nonzero means, they are independent of every component of the treatment index. The independent instrument noise v_{ji} has the same zero-covariance property. Expanding covariance by linearity, for $j = 1, 2$,

$$\begin{aligned} \operatorname{Cov}(Z_{ji}, d_i^*(\kappa)) = {} & \kappa \operatorname{Cov}(z_{ji}, z_{1i} + z_{2i}) + c(\kappa) \operatorname{Cov}(z_{ji}, w_i) + \operatorname{Cov}(z_{ji}, \epsilon_i) \\ & + \kappa \operatorname{Cov}(v_{ji}, z_{1i} + z_{2i}) + c(\kappa) \operatorname{Cov}(v_{ji}, w_i) + \operatorname{Cov}(v_{ji}, \epsilon_i) \\ & + \kappa \operatorname{Cov}(C_{ji}, z_{1i} + z_{2i}) + c(\kappa) \operatorname{Cov}(C_{ji}, w_i) + \operatorname{Cov}(C_{ji}, \epsilon_i). \end{aligned} \tag{B.5}$$

In the first line, the covariance with $z_{1i} + z_{2i}$ is one: z_{ji} has covariance one with itself and zero with the other primitive excluded component. Its covariances with w_i and ϵ_i are zero by independence. All three terms in the second line vanish because v_{ji} is independent of the index components. All three terms in the third line vanish because the controls are independent of those components. The nonzero means of the transformed controls do not change this independence argument. Only the first term remains, giving

$$\operatorname{Cov}(Z_{ji}, d_i^*(\kappa)) = \kappa, \qquad j = 1, 2.$$

This result concerns the latent index, not the nonlinear treatment: it does not assert that $\operatorname{Cov}(Z_{ji}, D_i(\kappa)) = \kappa$. Thus κ orders designed relevance, but is neither an F-statistic nor a universal measure of IVQR identification strength.

B.5 Preserving the Relationship Generating Endogeneity

The experiment should also preserve the relationship between treatment and the unobserved outcome disturbance. Expanding covariance with the index shows which component carries that relationship:

$$\begin{aligned} \operatorname{Cov}(u_i, d_i^*(\kappa)) &= \kappa \operatorname{Cov}(u_i, z_{1i} + z_{2i}) + c(\kappa) \operatorname{Cov}(u_i, w_i) + \operatorname{Cov}(u_i, \epsilon_i) \\ &= 0 + 0 + 0.3 = 0.3. \end{aligned}$$

The first term is zero because the primitive excluded components are independent of u_i. The second is zero because the new noise w_i is independently introduced. The third is the assumed baseline covariance 0.3 between u_i and ϵ_i; it is unchanged by scaling the other components.

To establish more than unchanged covariance, use joint normality. The pair (u_i, ϵ_i) is jointly normal and independent of the other normal primitives, so $(u_i, d_i^*(\kappa))$ is jointly normal as well. Both means are zero. The diagonal entries of its covariance matrix are the variance of u_i, one, and the index vari-

ance, three. Its two off-diagonal entries equal the covariance 0.3 just computed:

$$\begin{pmatrix} u_i \\ d_i^*(\kappa) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 3 \end{pmatrix}\right). \tag{B.6}$$

For a jointly normal pair, its mean vector and covariance matrix determine its distribution. Their invariance therefore establishes an unchanged joint distribution, a conclusion that covariance alone would not justify without joint normality. Applying Φ to the second component preserves the joint distribution of $(u_i, D_i(\kappa))$. The construction preserves the relationship generating endogeneity, not merely one covariance.

Treatment realizations can still differ across κ within the same replication because the signal and noise weights change. Relationships involving the instruments are deliberately changed. Reusing primitive draws across settings aids comparison, but is not the reason for marginal invariance: the distributional argument above holds separately at every setting.

B.6 Structural Effects and Endpoint Cases

The structural effect follows from the potential-outcome equation, Equation (40). For a fixed treatment value $d \in (0, 1)$,

$$Y_i(d) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d(1 + u_i).$$

The constant one inside $d(1 + u_i)$ is the baseline coefficient on treatment in the outcome equation; it is distinct from the outcome intercept of one. The coefficient five on each of the seven relevant controls is also inherited from the baseline. Since u_i is independent of X_i, its conditional τ-quantile is the standard-normal quantile $\Phi^{-1}(\tau)$. For $d > 0$, multiplying by d preserves order, so the quantile of $d(1 + u_i)$ is $d[1 + \Phi^{-1}(\tau)]$. Adding the fixed intercept and control contribution gives

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d\left[1 + \Phi^{-1}(\tau)\right],$$

and the coefficient of treatment in the structural quantile function is $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$. This is a potential-outcome quantile at fixed treatment, not an observational quantile conditional on the endogenous realized treatment. Modifying treatment assignment does not change the potential-outcome equation or the distribution of u_i conditional on X_i, and hence does not alter this structural effect.

At $\kappa = 1$, $c(1) = 0$ and $D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i)$, recovering the baseline. At the conceptual endpoint $\kappa = 0$,

$$d_i^*(0) = \sqrt{2}\, w_i + \epsilon_i, \qquad D_i(0) = \Phi(\sqrt{2}\, w_i + \epsilon_i),$$

so the primitive excluded signal disappears from assignment. At the weakest simulated setting, $\kappa = 0.10$, the signal variance is still positive: $2(0.10)^2 = 0.02$. These endpoint statements describe the construction; they are not a proof of global identification or non-identification for every IVQR model.

B.7 Numerical Illustration

Table B.1 extends the worked $\kappa = 0.50$ example to all four settings. Signal variance, replacement-noise variance and ϵ_i's unit variance always sum to three.

Table B.1. Variance allocation in the κ-extension.

κ	Coefficient $\sqrt{2(1-\kappa^2)}$	Signal variance $2\kappa^2$	Noise variance $2(1-\kappa^2)$	Total index variance
1.00	0.0000	2.0000	0.0000	3.0000
0.50	1.2247	0.5000	1.5000	3.0000
0.25	1.3693	0.1250	1.8750	3.0000
0.10	1.4071	0.0200	1.9800	3.0000

Note: Values are calculated directly from the construction, not simulated. The disturbance ϵ_i contributes the remaining variance of one in every row.

Appendix C Coverage Audit and Inferential Informativeness

C.1 Direct-at-Truth Coverage Audit

Coverage is evaluated directly at the exact true structural effect $\alpha_0(\tau)$, independently of whether that value lies on the finite candidate grid. The forensic audit checked 60,000 direct truth evaluations: 500 replications for each of 120 sample-size, quantile, instrument-strength, and estimator cells.

Wilson 95% Monte Carlo intervals exclude nominal 0.95 coverage in 54 of the 120 cells. The audit finds no missing or nonfinite true-parameter W statistics. The large coverage distortions reported in the main text are therefore neither plotting-grid artifacts nor consequences of missing-value handling. The frozen audit report and supporting diagnostics remain preserved in the repository.

These checks rule out those numerical explanations for the reported coverage patterns. They do not establish correct theoretical specification, validate the asymptotic reference distribution for this DGP, or establish nominal finite-sample calibration.

C.2 Coverage-Audit Summary

Table C.1 summarizes the audit by estimator and sample size. Each row contains 20 quantile–κ cells. Recalculation from integer acceptance counts reproduces 53 cells with an upper Wilson endpoint below 0.95 and one with a lower endpoint above 0.95, giving the reported total of 54. Comparisons use unrounded endpoints and the same 500-evaluation denominator in every cell.

Table C.1. Direct-at-truth coverage audit.

Estimator	n	Cells	Coverage min.	Coverage max.	Upper < 0.95	Lower > 0.95	Missing/ nonfinite
Oracle-GMM	500	20	0.858	0.954	6	0	0
Oracle-GMM	1000	20	0.860	0.964	8	0	0
Full-GMM	500	20	0.442	0.962	9	0	0
Full-GMM	1000	20	0.502	0.962	12	0	0
DML-IVQR	500	20	0.658	0.976	8	1	0
DML-IVQR	1000	20	0.498	0.968	10	0	0

Note: Each row covers 20 quantile–κ cells, with 500 direct-at-truth evaluations per cell. Coverage is reported as a proportion. The upper/lower columns count cells whose unrounded Wilson upper endpoint is below 0.95 or lower endpoint is above 0.95. Intervals use integer acceptance counts, the verified denominator, and the two-sided 95% Wilson formula without continuity correction. They are pointwise Monte Carlo intervals, not simultaneous or multiple-testing-adjusted intervals; exclusion counts are descriptive audit summaries. Missing/nonfinite counts concern direct-at-truth W evaluations (missing values are counted once).

C.3 Full-Domain Grid Acceptance

The domain-expansion diagnostics in Appendix A are complemented by Figure C.1, which reports how often every evaluated point in the primary domain is accepted. Reading this frequency alongside coverage distinguishes retention of the true effect from the ability to exclude competing candidates within the maintained domain. Coverage near nominal can coexist with frequent acceptance of all primary-grid candidates: retaining the true value does not imply excluding alternatives.

The heatmap uses stored grid-profile evaluations, whereas coverage uses a separate evaluation directly at truth. These are distinct computations; in DML-IVQR, fresh pivotal draws can differ between them, so their acceptance events need not coincide even when truth is a grid point. Full-domain grid acceptance refers only to the 41 evaluated points in $A_0 = [-1, 3]$, not to acceptance of the real line or global unboundedness of an unrestricted confidence region.

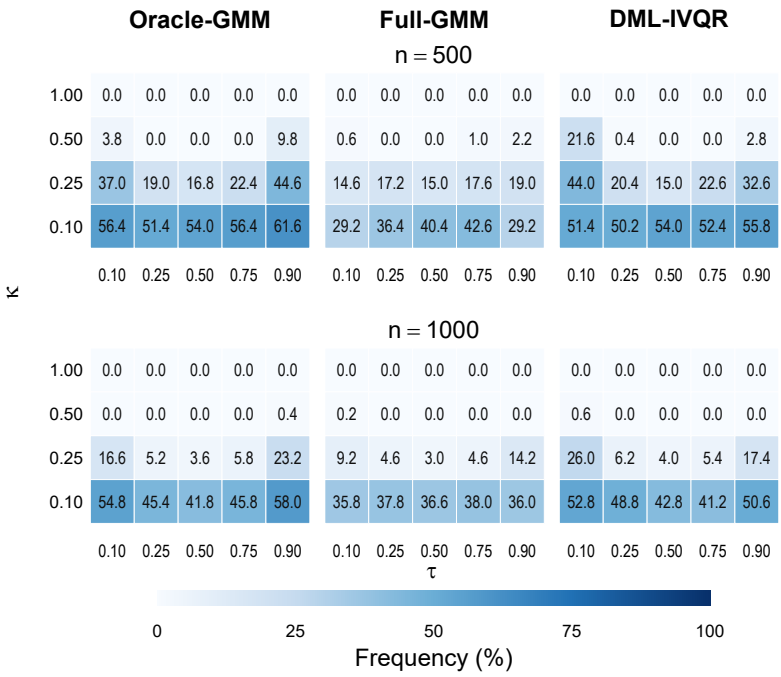


Figure C.1. Full-domain grid acceptance frequencies.

Note: Each cell reports the percentage of 500 complete replications accepting all 41 evaluated grid points in $A_0 = [-1, 3]$, with step size $h = 0.1$. Panel columns identify estimators; panel rows show $n = 500$ and $n = 1000$. Within each panel, columns show quantiles τ and rows show relevance κ, declining from 1.00 to 0.10. Darker shading on the shared 0–100% scale means more frequent full-domain grid acceptance. This event neither denotes acceptance of the entire real line nor establishes an unbounded unrestricted confidence region.

Appendix D Reproducibility and Implementation Notes

The repository for this thesis is publicly available and contains all the code needed to reproduce the reported results: https://github.com/U-Olim/DML-IVQR_Master_Thesis.git.

The preserved author-replication and main-extension environment is R 3.4.3 on 64-bit Windows. The main Monte Carlo uses $R = 500$ replications per design and seed 675. The grid-domain diagnostic also uses seed 675 and reuses saved primitives in its second stage. The separate instrument-strength calibration uses seed 20260813. The frozen reporting outputs are exactly reproducible from the archived CSVs. The parent-process seed records the simulation setup, but because the preserved parallel DML implementation does not explicitly initialize worker-specific RNG streams, exact regeneration of all pivotal penalty draws from the seed alone is not established.

The reporting layer reads frozen simulation, calibration, and diagnostic outputs without re-estimating models. Reporting consistency checks and source/output hash manifests connect those records to the numerical tables. Figure-data checks verify all 480 plotted values against the frozen reporting CSVs, while the figure manifest records source and final-image hashes. Full supplementary

tables, RMSE figures, detailed diagnostics, configuration files, and environment records remain in the repository even where they are not printed in the current PDF.

The 480-value figure check concerns the original archived figure set. The selected-figure validation additionally checks 120 coverage values, 120 full-domain acceptance frequencies, and 168 median rejection probabilities against final outputs. The appendix reporting step regenerates Table C.1 from saved direct-at-truth evaluations and checks the results against the frozen coverage audit. It also generates the analytical illustration in Table B.1 directly from the variance formulas. Neither step re-estimates models. The additional reporting outputs retain integer acceptance counts, unrounded Wilson endpoints, reconciliation checks, and input hashes.

An implementation-specific detail concerns the fitted-normal density proxy used in the DML-IVQR nuisance step. The preserved code calls `dnorm(e, mean(e), var(e))`. In R, the third argument of `dnorm` is interpreted as a standard deviation, whereas the preserved code supplies the residual sample variance. This behavior is retained to preserve the executable implementation used in the simulation study.

Declaration

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

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